Sources and attribution

Historical manuscript notes of the I. M. Gelfand Seminar. MCCME digital archive: Notes of the talks at the I. M. Gelfand Seminar.

MCCME identifies M. A. Shubin as the note-taker. The Clay Mathematics Institute funded the scanning. MCCME acknowledges M. A. Shubin, S. I. Gelfand, the Clay Mathematics Institute and Jim Carlson. MCCME archive notice.

Chapter 31. Manuscript frames 1385–1424. Catalogue snapshot: 5 October 2026.

Speakers named in the catalogue associations for the complete edition, using the catalogue's spelling:

I.M. Gelfand; R.L. Dobrushin; Irving Segal; Laurent Schwartz; S.G. Gindikin; A.Ya. Povzner; V.I. Arnold; A.A. Kirillov; I.I. Pyateski-Shapiro; M.I. Vishik; Yu.V. Linnik; E.A. Gorin; D.A. Kazhdan; V.P. Palamodov; A.S. Dynin; S.P. Novikov; Ya.G. Sinai; I.M. Gelfand, A.Ya. Povzner,Berezin; V.N. Latyshev; S.I. Gelfand; I.N. Bernstein; A.M. Stepin; R.A. Minlos; L.D. Faddeev; Paul Halmos; J.J. Kohn; Robert Finn; S.V. Smirnov; V.I. Matsaev; A.G. Kushnirenko; M.S. Agranovich; F.I. Karpelevich; V.A. Borovikov; V.B. Lidskii; G.E. Shilov; Lee Lorch; Louis Nirenberg; O.A. Oleinik; A.N. Parshin; D.B. Fuchs; S.G. Krein; Jean Leray; L.G. Makar-Limanov; A.Yu. Geronimus; Alan Weinstein ; M.L. Gromov; G.A. Margulis; Chandler Davis; V.M. Buchstaber; B.R. Vainberg; Gaetano Fikera; Dusa McDuff; Edwin Hewitt; B.V. Fedosov; A.M. Gabrielov; S. Rolevich; A.M. Perelomov; B.I. Rosenfeld; Vlastimil Ptak; Michael Atiyah; Pierre Deligne; A.M. Vinogradov; L.N. Vasserstein; Jack Morava; M.Sh. Birman; Yu.I. Manin; A.G. Kostyuchenko; V.I. Lomonosov; F.A. Berezin; G.M. Henkin; A.S. Mishchenko; A.N. Rudakov; A.I. Schnirelman; V.A. Ponomarev; Tony Springer; V.E. Zakharov; V.N. Tulovskii; A.B. Shabat; B.M. Gurevich; Yu. Koblenz; Enrico Guisti; Czeslaw Bessaga; B.A. Dubrovin; Graeme Segal; A.M. Polyakov; M.I. Graev; James King; Bert Kostant; E.S. Fradkin; I.M. Krichever; V.G. Drinfeld; M.V. Losik; A.V. Roiter; A.A. Migdal; V.N. Popov ; J.J. Duistermaat; A.M. Vershik; A.S. Tyurin; D.A. Kirzhnits; L.A. Dickey; S.M. Vishik; Bela Sekefalvi-Nagy; Kazuhiko Aomoto; Francesco Calogero; V.S. Kruskal; A.S. Schwarz; Alain Guichardet; A.A. Belavin; A.A. Beilinson; V.I. Ogievetskii; D.R. Lebedev; G.B. Shabat; Renata Kallosh; Raphael Hoegh-Krohn; I.A. Batalin; I.Yu. Kobzarev; A.V. Zelevinsky; Robert MacPherson; G.I. Olshanskii; A.B. Goncharov; Bernard Gaveau; A.G. Khovansky; I.V. Cherednik; B.Ya. Kazarnovskii; P.G. Grinevich; Jean-Luc Brylinski; Eckart Viehweg; Christian Kassel; A.V. Bobylev; B.V. Lidskii; Benedict Seifert; A.G. Elashvili; Werner Mueller; V.L. Ginzburg; John Milnor; Andre Haefliger; A.B. Givental; Mariusz Wodzicki; Al-dr.B. Zamolodchikov; Jurgen Frohlich; B.L. Feigin; Yu.A. Neretin; M.A. Semenov-Tyan-Shanski; George Wilson; Yu.L. Daletskii; V.A. Chulaevskii; P.B. Wiegman; F. Nikiforov; Yu.M. Berezanskii; D.A. Leites; Ivan Penkov; Jean-Pierre Serre; M.M. Kapranov; Marie-France Vigneras; Yu.S. Ryzhikova; V.G. Knizhnik; M.L. Kontsevich; A.Yu. Morozov; A.N. Varchenko; Yan I. Kogan; V.A. Kazakov; V.Yu. Ovsienko; I.S. Zakharevich; Don Zagier; D.V. Alekseevskii; A.P. Veselov; Klyachko; A.O. Radul; N.Yu. Reshetikhin; A.G. Usheridze; Gregory Moore; V.S. Dotsenko.

A catalogue association does not by itself establish an exact talk boundary within a manuscript page or identify the writer of every note.

The modern LaTeX text is AI-transcribed. The English edition includes AI-translated passages and AI-transcribed passages originally written in English. Diagram reconstructions are AI-created and typesetting is AI-produced. Historical lecture and manuscript authorship remains with its human authors.

Rights in the original notes, scans and other third-party material remain with their authors and respective rights holders, including heirs. To the extent that the project holds copyright or related rights in its own contributions, it dedicates those rights under CC0 1.0 Universal. The dedication applies only to rights held by the project.

These editions are freely available and are published solely for non-commercial educational purposes. If a rights holder or their representative objects to publication, please identify the affected material in GitHub Issues so we can take it down.

Exact image URLs, file digests and catalogue associations accompany the editable sources.

Chapter 31

GELFAND-P001385

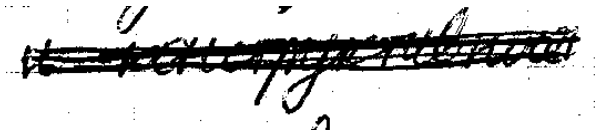
[source number: 1]

5/X-87 Gelfand Seminar.

A. Beilinson. Constructible sheaves (continued).

Syllabus on sheaves.

1. Sheaf, category of sheaves.
2. Operations on sheaves:
   a) Gluing over an open cover
   b) Gluing for an open and a closed set
   c) Direct and inverse images.
3. Locally constant sheaves.
4. Combinatorial [crossed out: possibly “constructible”[?]] sheaves = representations of ordered sets.



Uncertain reading. possibly “constructible”

Original source details: GELFAND-P001385

5. Constructible sheaves.

6. Cohomology with coefficients in sheaves.

7. Poincaré–Verdier duality

For items 1–5, see the preceding notebook.

Sheaves form an abelian category.

Let $\mathcal{F}, G$ — be sheaves, with a morphism $\mathcal{F} \xrightarrow{\varphi} G$.

All morphisms form a vector space (over $\mathbb{C}$)

Definition 1.

Ker $\varphi \subset \mathcal{F}$ [Below the formula: subsheaf]

$$(\ker \varphi)(V) = \{a \in \mathcal{F}(V),\ \varphi(a) = 0\}$$

Coker φ — the sheaf associated to the presheaf C given by

$$C(V) = \text{Coker}\big(\varphi : \mathcal{F}(V) \longrightarrow G(V)\big)$$

Constructing a sheaf from a presheaf.

Let C — be a presheaf.

Let V — be an open set, $V = \bigcup V_\alpha$ and suppose we are given $a_\alpha \in C(V_\alpha)$, compatible with one another:

$$a_\alpha|_{V_\alpha \cap V_\beta} = a_\beta|_{V_\alpha \cap V_\beta}.$$

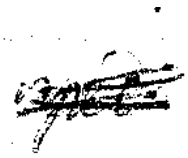
Such a family determines a section of the sheaf. There is an identification of sections "obtained"? from different families.



Uncertain reading. "obtained"

Original source details: GELFAND-P001385

C^+ — [crossed out: possibly «пред.» ("pre...")?] the sheaf associated to C.



Uncertain reading. possibly «пред.» ("pre...")

Original source details: GELFAND-P001385

GELFAND-P001386

There is a map $C \longrightarrow C^+$, universal in the following sense

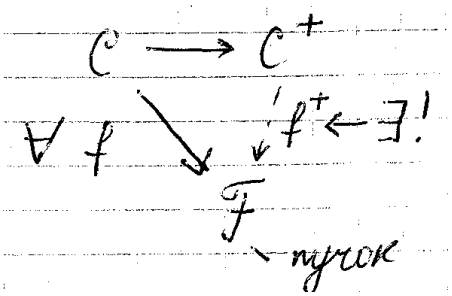
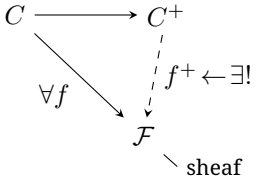


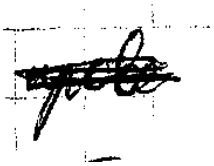
Diagram and typed description.



Original source details: GELFAND-P001386

$\forall$ map $f: C \longrightarrow \mathcal{F}$ [below $\mathcal{F}$: sheaf] $\exists!$ $f^+: C^+ \longrightarrow \mathcal{F}$, such that the diagram commutes

Exercise 1. [crossed out: "про..."?] construct C^+.



Uncertain reading. "про..."?

Original source details: GELFAND-P001386

There are maps

$$\operatorname{Ker}\varphi \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\varphi} G \xrightarrow{\beta} \operatorname{Coker}\varphi$$

Exercise 2. $\operatorname{Ker}\beta = \operatorname{Coker}\alpha$

An exact sequence of sheaves is defined in the natural way.

Exercise 3. The sequence $\mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3$ is exact $\Longleftrightarrow$ $\forall x \in X$ the sequence $\mathcal{F}_{1x} \longrightarrow \mathcal{F}_{2x} \longrightarrow \mathcal{F}_{3x}$ is exact.

(Consider the inclusion $x \xrightarrow{i_x} X$, then the stalk $\mathcal{F}_x$ at the point x has the form $\mathcal{F}_x = i_x^*\mathcal{F}$)

Definition 2 (Flabby sheaf.)

A sheaf $\mathcal{F}$ is called flabby if any section over an open set V extends to a section over all of X, i.e. $\forall$ open $V \subset X$

$$\mathcal{F}(X) \longrightarrow \mathcal{F}(V)$$

is an epimorphism.

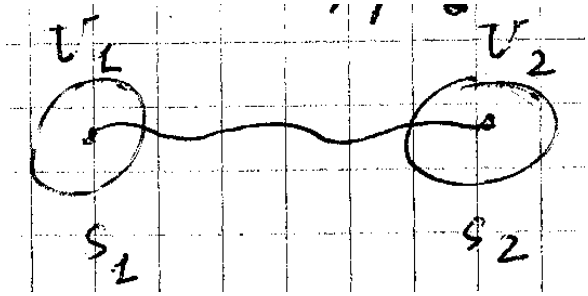
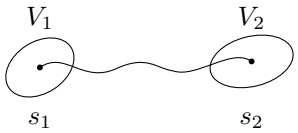


Diagram and typed description.



Original source details: GELFAND-P001386

In locally constant sheaves, sections can be transported [inserted above the line: (along a path)] and therefore such sheaves are not flabby.

Problem 4. Γ — a [inserted above the line: finite] ordered set; introduce the topology

$$\overline{x} := \{y : \ y \leq x\} \quad \text{— closure of the point } x$$

a) Sheaves on Γ = representations of Γ

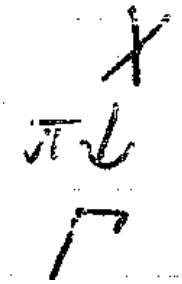
b) Describe the flabby sheaves on Γ

GELFAND-P001387

[source number: 2]

c) Connection with combinatorial sheaves

Suppose X has a stratification, consider the map



Original source details: GELFAND-P001387

Diagram and typed description.

$$\begin{array}{c} X \\ \pi \downarrow \\ \Gamma \end{array}$$

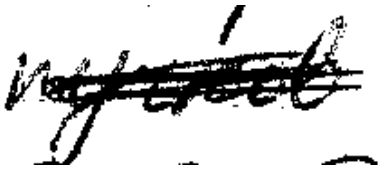
(to the point x — its stratum)

Γ — ordered set of strata.

[below Γ:] Give Γ the (Zariski) topology

Claim.

$\forall$ combinatorial sheaf on X is the inverse image of [crossed out: “пучков” (of sheaves)?] a sheaf on Γ (and conversely)



Original source details: GELFAND-P001387

Uncertain reading. “пучков” (of sheaves)?

$$\pi^*: \text{sheaves on } \Gamma \xrightarrow{\sim} \text{combinatorial sheaves on } X.$$

Exercise 5. a) Suppose

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is a short exact sequence of sheaves. Then the sequence of sections

$$0 \longrightarrow \mathcal{F}_1(X) \longrightarrow \mathcal{F}_2(X) \longrightarrow \mathcal{F}_3(X)$$

is exact.

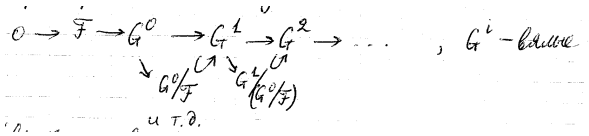
b) If $\mathcal{F}_1$ is flabby, the following sequence is exact

$$0 \longrightarrow \mathcal{F}_1(X) \longrightarrow \mathcal{F}_2(X) \longrightarrow \mathcal{F}_3(X) \longrightarrow 0$$

Flabby resolution of a sheaf.

Exercise 6 Let $\mathcal{F}$ be a sheaf. Then 1) It can be embedded in a flabby sheaf $\mathcal{F} \hookrightarrow G$.

2) Now construct from $\mathcal{F}$ an exact sequence



Original source details: GELFAND-P001387

Diagram and typed description.

$$0 \longrightarrow \mathcal{F} \longrightarrow G^0 \longrightarrow G^1 \longrightarrow G^2 \longrightarrow \cdots, \quad G^i \text{ — flabby}$$

$$G^0/\mathcal{F} \qquad G^1/(G^0/\mathcal{F})$$

and so on

(flabby resolution)

3) $\forall$ two flabby resolutions can be embedded in a common third one.

Definition 3.

$$H^i(X, \mathcal{F}) = H^i\big(0 \longrightarrow G^0(X) \longrightarrow G^1(X) \longrightarrow \ldots\big)$$

(here $0 \longrightarrow \mathcal{F} \longrightarrow G^0 \longrightarrow G^1 \longrightarrow \ldots$ — flabby resolution $\mathcal{F}$)

On the right is the cohomology of a complex of vector spaces.

Definition 4. [crossed out: Dual sheaf.] Cohomology with compact support.

Suppose X is locally compact. Then we can consider sections with compact support: $\mathcal{F}_c(X)$ — sections with compact support on X.

GELFAND-P001388

$$H^i_c(X, \mathcal{F}) = H^i\big(0 \longrightarrow G^0_c(X) \longrightarrow G^1_c(X) \longrightarrow \ldots\big)$$

Definition 5. Dual sheaf.

Presheaf: $\mathcal{F}^*(V) = \mathcal{F}_c(V)^*$ (analogue: “distribution” or possibly “distributions”[?])



Uncertain reading. “distribution” or possibly “distributions”

Original source details: GELFAND-P001388

(dual space in the sense of linear algebra)

$$V_1 \subset V_2 \;\Rightarrow\; \mathcal{F}_c(V_1) \longrightarrow \mathcal{F}_c(V_2) \quad \text{(embedding)}$$

and this gives $\mathcal{F}^*(V_2) \longrightarrow \mathcal{F}^*(V_1)$

Exercise 7.

a) $\mathcal{F}^*$ is a sheaf if $\mathcal{F}$ is flabby.

b) If $\mathcal{F}^*$ is a sheaf, then $\mathcal{F}^*$ is flabby, i.e. $\forall \underset{\text{open}}{V} \subset X$ the map $\mathcal{F}^*(X) \longrightarrow \mathcal{F}^*(V)$ is surjective

GELFAND-P001389

12 October 1987. Gelfand Seminar.

Vakhtangov told the actors in Princess Turandot to come on stage in fashionable clothes and change at once (put on a turban, etc.). The themes of mathematics are eternal, only the language changes.

I. M. G. On a combinatorial conjecture of Dyson, connected with polynomial coefficients

A paper from the thick green collection on combinatorics

Consider

$$F(x,a)=\prod_{i\neq j}\left(1-\frac{x_j}{x_i}\right)^{a_j},\qquad a_j\in\mathbb{N}. \tag{1}$$

Find the constant term

Dyson's conjecture (1962), the answer

$$\frac{(a_1+a_2\ldots+a_n)!}{a_1!a_2!\ldots a_n!} \tag{2}$$

In 1962 two proofs were given in J. Math. Phys. (long ones)

In 1970 in J. Math. Phys., a proof by

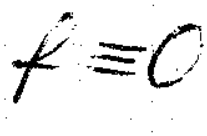


Uncertain reading. the name appears as "Вуда" (Vuda)

Original source details: GELFAND-P001389

in two lines.

Proof Applying Lagrange's interpolation formula to the function $f\equiv 0$



Uncertain reading. The manuscript reads $f\equiv 0$.

Original source details: GELFAND-P001389

and then setting $x=0$, we obtain

$$\sum_j\prod_{i\neq j}\left(1-\frac{x_j}{x_i}\right)^{-1}=1 \tag{3}$$

Multiplying by $F(x,a)$, we obtain

$$\sum_j F(x, a_1, \ldots, a_{j-1}, a_j - 1, a_{j+1}, \ldots, a_n) = F(x, a) \tag{4}$$

[crossed out: illegible opening mark?]



Uncertain reading. illegible opening mark

Original source details: GELFAND-P001390

Let $G(a)$ be the constant term. Then from (4)

$$G(a) = \sum G(a_1, \ldots, a_{j-1}, a_j - 1, a_{j+1}, \ldots, a_n), \tag{5}$$

which gives a recurrence formula.

When $a_j = 0$ we obtain

$$G(a_1, \ldots, a_{j-1}, 0, a_{j+1}, \ldots, a_n) = G(a_1, \ldots, a_{j-1}, a_{j+1}, \ldots, a_n) \tag{6}$$

Moreover,

$$G(0) = 1 \tag{7}$$

The relations (5)–(7) are sufficient as recurrence relations. They determine everything and yield (2).

A. Veselov

On geodesics on an ellipsoid and motion on a sphere

Following Jacobi and others, 1982. Knörrer's observation: equivalence of two problems.

[Two columns separated by a curved line. Left column.]

1) $(Aq, q) = 1$

geodesics on the ellipsoid

Weierstrass found them in θ-functions (1861)

Equation of the geodesics in the natural parameter:

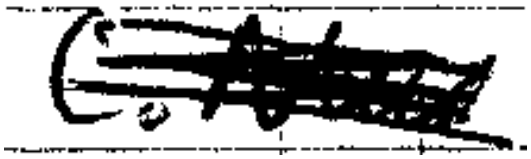
$$\ddot{q} = \lambda A q \tag{1}$$

(acceleration $\sim$ the normal)

[At right:] (λ is the Lagrange multiplier, found from the condition $(Aq, q) = 1$)

[Right column.]

2) Another problem [crossed out: C. the name after the initial is struck through, possibly "Neu..."?]



Uncertain reading. C. the name after the initial is struck through, possibly "Neu..."

Original source details: GELFAND-P001390

C. Neumann (1859)

Motion with potential energy

$$U(x) = \frac{1}{2}(Ax, x)$$

on the sphere $|x|^2 = 1$

Here the equation is

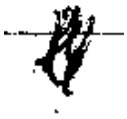
$$x'' = -Ax + ux \tag{1'}$$

or in coordinates

$$x_i'' = -a_i x_i + u x_i \tag{1''}$$

[a symbol is crossed out at right: q? / u?[?]]

Uncertain reading. q? / u?



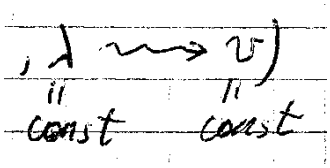
Original source details: GELFAND-P001390

[Below the columns.]

Switch to another parameter

$$q'' = -Aq + vq' \tag{2}$$

(the changes can be written out explicitly, $\underset{\|}{\lambda} \rightsquigarrow \underset{\|}{v}$ $\underset{\text{const}}{\quad}$ $\underset{\text{const}}{\quad}$)



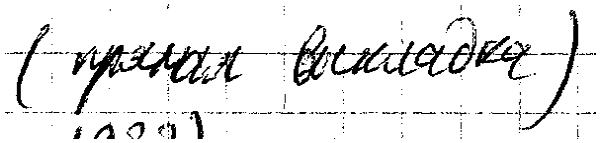
Uncertain reading. The manuscript has “const” under λ and v.

Original source details: GELFAND-P001390

and then the substitution

$$x = \frac{Aq}{|Aq|} \qquad \text{(Gauss substitution)} \tag{3}$$

takes solutions of problem 1) to solutions of problem 2) (direct computation)



Uncertain reading. The manuscript reads “прямая выкладка” (direct computation).

Original source details: GELFAND-P001390

(H. Knörrer, 1982)

Equation

[At right:] Introduce the operator:

$$L = -\frac{d^2}{dt^2} + u, \qquad Lx_i = a_i x_i \tag{4}$$

(x_i — eigenfunction!!)

GELFAND-P001391

[source number: 12]

At left:

$$\widetilde{L} = -\frac{d^2}{dt^2} + v\frac{d}{dt} \tag{5}$$

(conjugate to L by the Gauss substitution or the Miura substitution)

Knörrer arrived at it through the Moser–Trubowitz isomorphism

Consider problem 2)

There are two symplectic forms.

Problem (I. M. G.) There is a symmetric matrix A and a matrix K of rank one. How, from the eigenvalues of $A + K$ can one recover K? (A is fixed)

— this is exactly Jacobi's formula.

V. For $n = 2$ introduce new coordinates

$$M = [\dot{x}, x], \qquad \text{(angular momentum — cross product)}$$
$$p = x$$

1°. Lie–Berezin–Kirillov bracket

$$\{M_i, M_j\} = \varepsilon_{ijk} M_k$$
$$\{M_i, p_j\} = \varepsilon_{ijk} p_k$$
$$\{p_i, p_j\} = 0$$

Hamiltonian

$$2H = M^2 + (Ap, p)$$

Then

$$\begin{cases} \dot{M}_i = \{M_i, H\} \\ \dot{p}_i = \{p_i, H\} \end{cases} \qquad \text{— the Neumann equations}$$

Second bracket

$$\varepsilon_{ijk} a_k M_k = \{M_i, M_j\}$$
$$\varepsilon_{ijk} a_k p_k = \{M_i, p_j\}$$
$$\varepsilon_{ijk} M_k = \{p_i, p_j\}$$

Hamiltonian $F = \frac{1}{2}p^2$

I. M. G. This agrees with what Arnold discussed

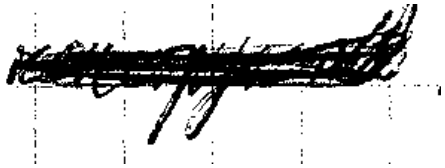
(see the previous notebook)

The system is Hamiltonian in both.

GELFAND-P001392

A. Beilinson. Constructible sheaves (continued)

Representations of a partially ordered set $\Longleftrightarrow$ [crossed out: possibly "constructible"?] combinatorial sheaves



Original source details: GELFAND-P001392

Uncertain reading. possibly "constructible"



Diagram and typed description.

$* \leq x$ $\qquad$ $\bullet \leq \bullet \geq \bullet$

Original source details: GELFAND-P001392

A combinatorial sheaf is a sheaf constant on each stratum.

Homological algebra.

$\mathcal{A}$ — abelian category (for example, of sheaves over a given space or of modules)

$C(\mathcal{A})$ — category of complexes over $\mathcal{A}$

one needs to know the derived category (Verdier, 1961)

Objects of [crossed out: possibly “из” (from) or a membership sign[?]] $C(\mathcal{A})$



Uncertain reading. possibly “из” (from) or a membership sign

Original source details: GELFAND-P001392

$$\ldots \longrightarrow X^i \xrightarrow{d} X^{i+1} \xrightarrow{d} X^{i+2} \longrightarrow \ldots$$
$$d^2 = 0$$

morphisms are morphisms of complexes (families of morphisms

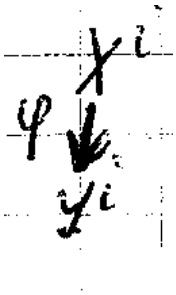


Diagram and typed description.

$$\begin{array}{c} X^i \\ \varphi \downarrow \\ Y^i \end{array}$$

Original source details: GELFAND-P001392

, commuting with d)

$C(\mathcal{A})$ — also an abelian category

Bounded complexes (only finitely many $\neq 0$) $C^b(\mathcal{A})$

bounded on the left or the right $C^{\pm}(\mathcal{A})$

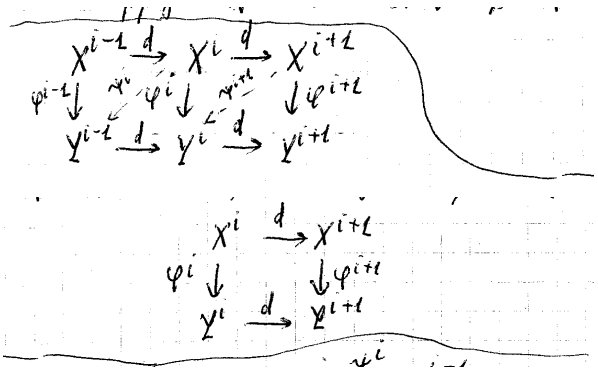
$$C^+(\mathcal{A}) \text{ — bounded on the left}$$
$$C^-(\mathcal{A}) \text{ — bounded on the right}$$

Homotopy category $K(\mathcal{A})$

the objects of $K(\mathcal{A})$ are the same as in $C(\mathcal{A})$, and the morphisms are equivalence classes of morphisms of $C(\mathcal{A})$ with respect to homotopy.

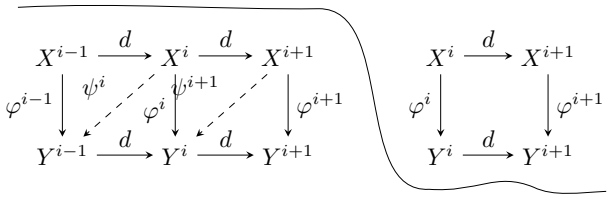
$$\mathrm{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet)$$

A morphism $X^\bullet \xrightarrow{\varphi} Y^\bullet$ is a [inserted above the line: such] family $\varphi^i \colon X^i \longrightarrow Y^i$, for which the diagram commutes:



Original source details: GELFAND-P001392

Diagram and typed description.



φ is called null-homotopic, if $\exists$ a family of maps $X^i \xrightarrow{\psi^i} Y^{i-1}$, such that

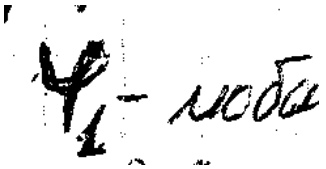
$$\varphi^i = d\psi^i + \psi^{i+1}d \tag{1}$$

GELFAND-P001393

[source number: 13]

Exercise 1. Check that for any system of maps ψ^i the family (1) gives a morphism of complexes, i.e. $d\varphi_i = \varphi_{i+1}d$.

Exercise 2. Morphisms homotopic to 0 form an ideal, i.e. if φ is homotopic to 0 and ψ_1 is any morphism,



Original source details: GELFAND-P001393

Uncertain reading. Here the symbol resembles ψ_1, and in the composition the symbol is φ_1.

then $\varphi \circ \varphi_1$ is homotopic to 0.
(and the same is true for the composition $\varphi_2 \circ \varphi$ with any morphism φ_2)

$$\mathrm{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet) = \mathrm{Hom}_{C(\mathcal{A})}(X^\bullet, Y^\bullet) \Big/ (\text{morphisms homotopic to } 0)$$

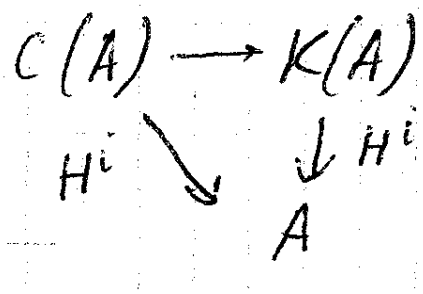
There are obvious functors

$$\mathcal{A} \longrightarrow C(\mathcal{A}) \longrightarrow K(\mathcal{A})$$

Cohomology functor

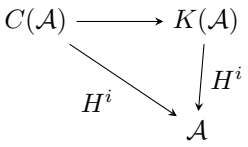
$$H^i \colon C(\mathcal{A}) \longrightarrow \mathcal{A} \qquad \left(H^i(X^\bullet) = \frac{\ker(d \colon X^i \longrightarrow X^{i+1})}{\mathrm{Im}(d \colon X^{i-1} \longrightarrow X^i)} \right)$$

Exercise 3. Prove that $\exists!$ a functor $H^i \colon K(\mathcal{A}) \longrightarrow \mathcal{A}$, such that



Original source details: GELFAND-P001393

Diagram and typed description.



On morphisms H^i is also defined in the obvious way.

Definition.

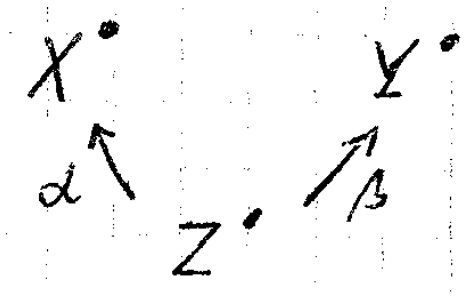
$$\varphi \colon X^\bullet \underset{\text{in the category } K(\mathcal{A})}{\longrightarrow} Y^\bullet$$

is called a quasi-isomorphism if it induces an isomorphism

$$H^i(X^\bullet) \longrightarrow H^i(Y^\bullet).$$

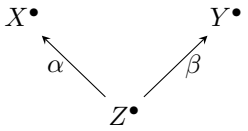
Derived category

Exercise 4. Suppose there are three complexes



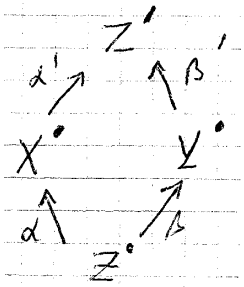
Original source details: GELFAND-P001393

Diagram and typed description.



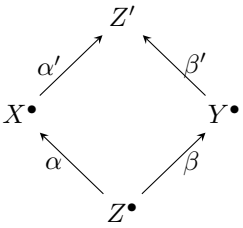
and morphisms, with α — a quasi-isomorphism.

Then there is a diagram (commutative)

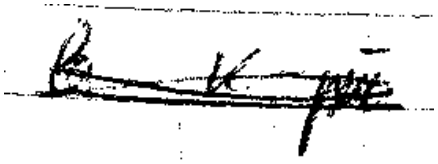


Original source details: GELFAND-P001393

Diagram and typed description.



[crossed out to the left of the diagram: “in which”?[?].]

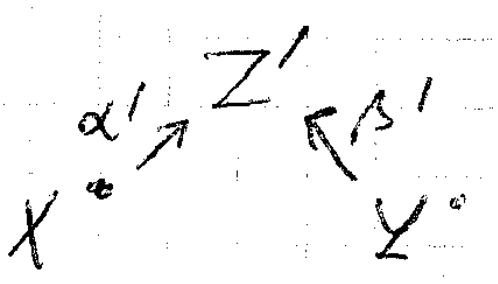


Original source details: GELFAND-P001393

Uncertain reading. “in which”?

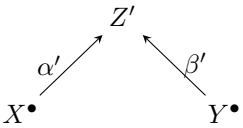
in which β' — a quasi-isomorphism.

The same is true if we start with the diagram



Original source details: GELFAND-P001393

Diagram and typed description.



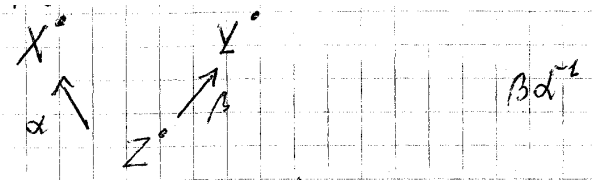
with quasi-isomorphism α'.

GELFAND-P001394

Derived category $D(\mathcal{A})$

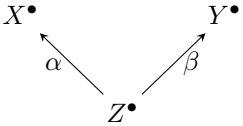
Objects = objects of $K(\mathcal{A})$ (i.e. complexes)

Morphisms [inserted above the line: $X^\bullet \longrightarrow Y^\bullet$] — equivalence classes of diagrams



Original source details: GELFAND-P001394

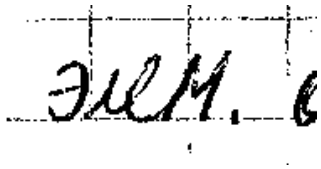
Diagram and typed description.



$\beta\alpha^{-1}$

(α — a quasi-isomorphism)

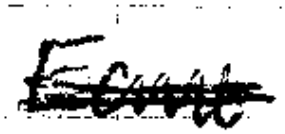
under the equivalence relation generated by the elementary relation



Uncertain reading. The abbreviation reads “элм.” (elementary).

Original source details: GELFAND-P001394

[crossed out: “There is”? / “If”?[?].]



Uncertain reading. “There is”? / “If”?

Original source details: GELFAND-P001394

(There is a functor $K(\mathcal{A}) \longrightarrow D(\mathcal{A})$)

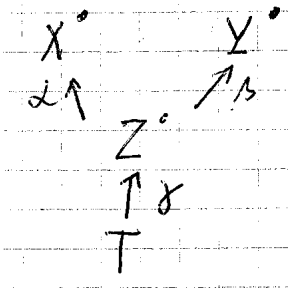
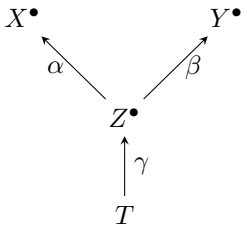
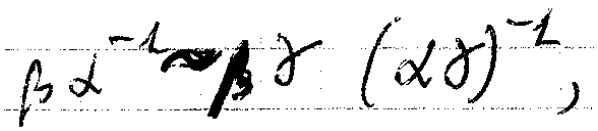


Diagram and typed description.



Original source details: GELFAND-P001394

$\beta\alpha^{-1} \sim \beta\gamma(\alpha\gamma)^{-1}$, where γ — a quasi-isomorphism.



Uncertain reading. The manuscript has reinforced strokes on $\sim$ and β.

Original source details: GELFAND-P001394

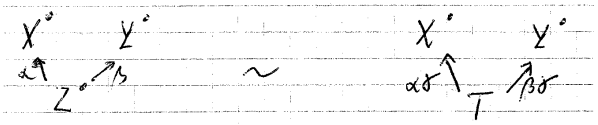
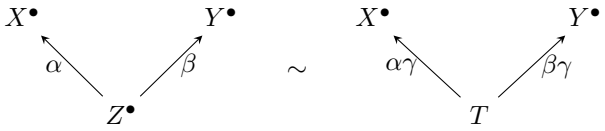


Diagram and typed description.



Original source details: GELFAND-P001394

We can now compose the morphisms.

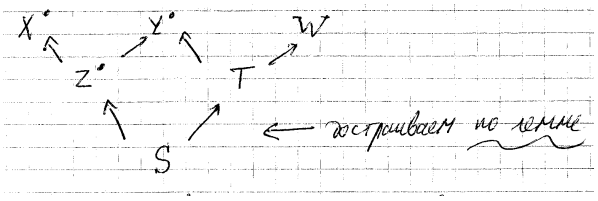
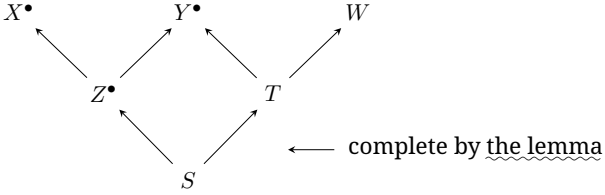


Diagram and typed description.



Original source details: GELFAND-P001394

the composition is the outside of the diagram.

GELFAND-P001395

19.10.87. Gelfand Seminar.

I. M. Gelfand Calculation of sums of kth powers and other sums



Uncertain reading. (following a recent idea?)

Original source details: GELFAND-P001395

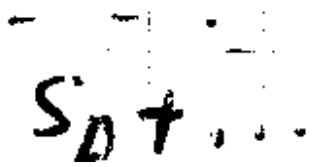
$$F'(x) = f(x) \qquad f(x) = x^k, \qquad F(x) = \frac{x^{k+1}}{k+1}$$

$$F(2) - F(1) = \frac{f(1)}{1!} + \frac{f'(1)}{2!} + \frac{f''(1)}{3!} + \dots$$
$$F(3) - F(2) = \frac{f(2)}{1!} + \frac{f'(2)}{2!} + \frac{f''(2)}{3!} + \dots$$
$$F(n+1) - F(n) = f(n) + \frac{f'(n)}{2!} + \frac{f''(n)}{3!} + \dots$$

$$F(n+1) - F(1) = S_0 + \frac{S_1}{2!} + \frac{S_2}{3!} + \dots$$

similarly,

$$f(n+1) - f(1) = S_1 + \frac{S_2}{2!} + \dots$$
$$f'(n+1) - f'(1) = S_2 + \frac{S_3}{2!} + \dots$$
$$\dots \qquad \dots$$
$$f^{(p)}(n+1) - f^{(p)}(1) = \dots \quad S_p + \dots$$



Uncertain reading. The index on the last line reads as S_p.

Original source details: GELFAND-P001395

[To the right of the system] Here

$$S_0 = f(1) + f(2) + \ldots + f(n)$$
$$S_1 = f'(1) + f'(2) + \ldots + f'(n)$$
$$\ldots$$
$$S_k = f^{(k)}(1) + f^{(k)}(2) + \ldots + f^{(k)}(n)$$

Hence [crossed out to the left of the determinant: “det”?[?]]



Uncertain reading. “det”?

Original source details: GELFAND-P001395

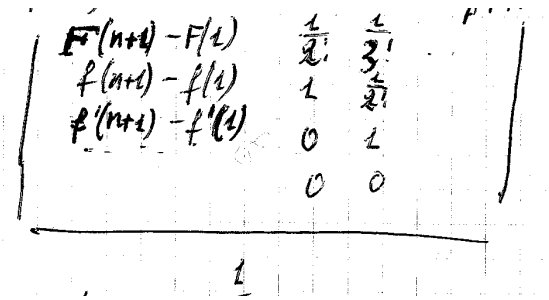


Diagram and typed description.

$$S_0 = \frac{\begin{vmatrix} F(n+1)-F(1) & \frac{1}{2!} & \frac{1}{3!} & \ldots \\ f(n+1)-f(1) & 1 & \frac{1}{2!} & \\ f'(n+1)-f'(1) & 0 & 1 & \\ \ldots & 0 & 0 & \end{vmatrix}}{1}$$

Original source details: GELFAND-P001395

Expand along the first column

$$S_0 = A_0\big(F(n+1) - F(1)\big) + A_1\big(f(n+1) - f(1)\big) + \ldots$$

coefficients are universal, it works for $\forall$ functions!!

Substitute $f(x) = e^{tx}$

banded determinants appear!

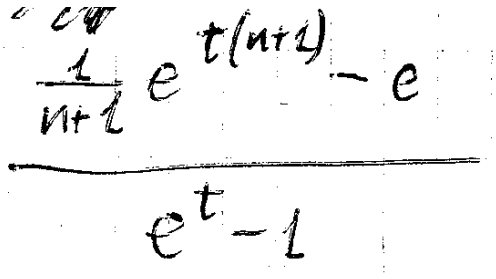
[crossed out: possibly $S_0 = A_0$[?]]



Uncertain reading. possibly $S_0 = A_0$

Original source details: GELFAND-P001395

$$\frac{\frac{1}{n+1}e^{t(n+1)} - e}{e^t - 1}$$



Uncertain reading. In the manuscript the numerator reads $\frac{1}{n+1}e^{t(n+1)} - e$.

Original source details: GELFAND-P001395

expand [crossed out: “series”?[?]]

Uncertain reading. “series”?



Original source details: GELFAND-P001395

in $t \implies$ Bernoulli numbers

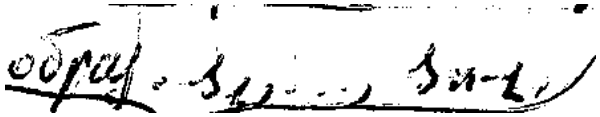
Problem. The symmetric group S_n is defined



Uncertain reading. The verb appears to read “определяется” (is defined).

Original source details: GELFAND-P001395

[inserted above the line: (generators $s_1, \ldots, s_{n-1}$)]



Uncertain reading. The abbreviation reads “образ.” (generators).

Original source details: GELFAND-P001395

by the relations

$$s_i^2 = 1, \qquad s_i s_j = s_j s_i \quad \text{when } |i-j| \geq 2, \qquad (s_i s_{i+1})^3 = 1$$

Consider the parabolic subgroup

$$G = \Sigma_n, \qquad H = \Sigma_k \times \Sigma_{n-k}, \qquad G \supset H$$

$$\mathcal{A} = \mathbb{C}[G] = \left\{ \sum_{g \in G} a_g g, \quad a_g \in \mathbb{C} \right\}$$

GELFAND-P001396

[source number: 23]

$$\mathcal{A} \supset \underset{\text{subalgebra}}{\mathcal{A}'} = \{x \in A,\ xh = hx = x \text{ for any } h \in H\}$$

[crossed out: illegible mark?]



Uncertain reading. illegible mark

Original source details: GELFAND-P001396

Beilinson, A. (continued)

Derived categories.

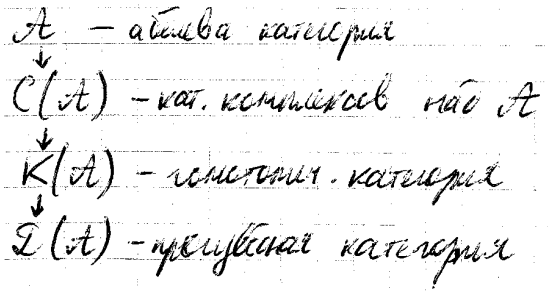


Diagram and typed description.

$\mathcal{A}$ — abelian category

$\downarrow$

$C(\mathcal{A})$ — category of complexes over $\mathcal{A}$

$\downarrow$

$K(\mathcal{A})$ — homotopy category

$\downarrow$

$D(\mathcal{A})$ — derived category

Original source details: GELFAND-P001396

the objects of $D(\mathcal{A})$ are complexes, and the morphisms are equivalence classes of diagrams

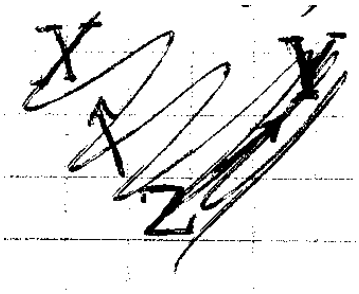
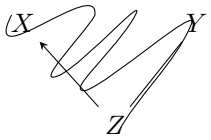


Diagram and typed description.



The diagram is crossed out. The right arrow is partly obscured.

Original source details: GELFAND-P001396

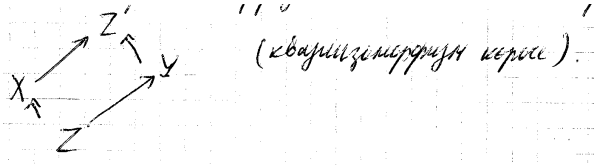
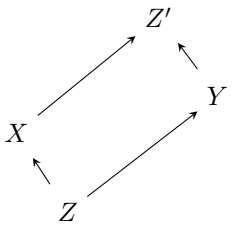


Diagram and typed description.



(quasi-isomorphisms are shorter).

Original source details: GELFAND-P001396

Example 1. Let X be a complex with [crossed out: possibly "co" (with)?] $H^i(X) = 0$ for $i \neq 0$. Then X is canonically quasi-isomorphic to $H^0(X)$:



Original source details: GELFAND-P001396

Uncertain reading. possibly "co" (with)

$$X \cong H^0(X) \text{ in } D(\mathcal{A})$$

(In the homotopy category this is not so!)

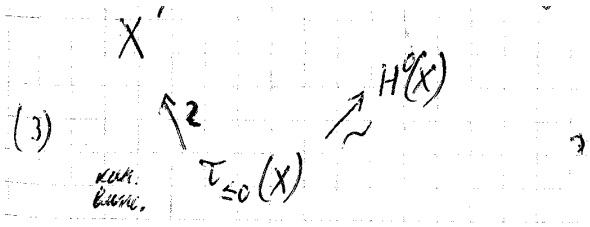
For a complex X introduce the subcomplex $\tau_{\leq 0}(X) \in X$:

$$\tau_{\leq 0}(X)^i = \begin{cases} X^i, & i < 0 \\ \ker(d\colon X^0 \to X^1), & i = 0 \\ 0, & i > 0 \end{cases} \tag{1}$$

It is easy to see that

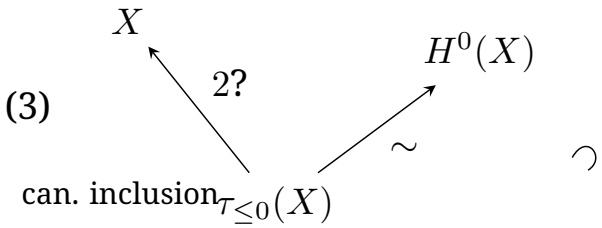
$$H^j(\tau_{\leq 0}(X)) = \begin{cases} H^j(X), & j \leq 0 \\ 0, & j > 0 \end{cases} \tag{2}$$

The diagram



Original source details: GELFAND-P001396

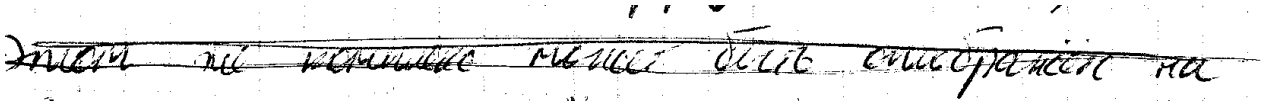
Diagram and typed description.



The mark by the left arrow resembles a 2 or a rotated tilde.

gives the required isomorphism in $D(\mathcal{A})$.

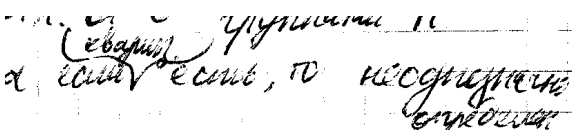
[crossed out: This same complex can be projected onto...?]



Original source details: GELFAND-P001396

Uncertain reading. This same complex can be projected onto...?

Remark. a complex X with several nonzero cohomology groups H^i is not quasi-isomorphic (in general) to the sum, and if there [inserted above the line: (quasi)] is one, the way is non-unique.



Original source details: GELFAND-P001396

Uncertain reading. The ending probably reads "in a non-unique way".

GELFAND-P001397

Structure of a triangulated category

$K(A)$ and $D(A)$ are not abelian categories.

Exercise 1. Check that

$$K(A) \text{ — abelian} \quad \Longleftrightarrow \quad \mathcal{A} \text{ is semisimple}$$

($\mathcal{A}$ — semisimple $\Longleftrightarrow$ [($\forall$ object) = (direct sum of irreducibles)]).

Cone of a morphism.

Let $f\colon X \longrightarrow Y$ be a morphism of complexes

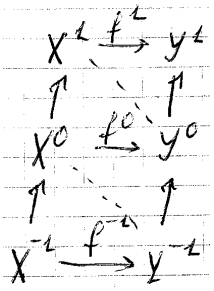
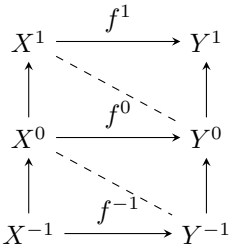


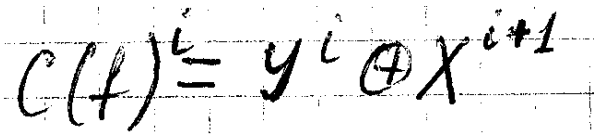
Diagram and typed description.



Original source details: GELFAND-P001397

[At right] consider all this as a bicomplex, i.e. write the complex

$$C(f)^i = Y^i \oplus X^{i+1}$$



Uncertain reading. The manuscript orders the summands as $Y^i \oplus X^{i+1}$.

Original source details: GELFAND-P001397

$$d(x, y) = (-d(x), d(y) + f(x))$$

There is a sequence

$$C(f[-1]) \longrightarrow X \xrightarrow{f} Y \underset{\text{incl.}}{\overset{g}{\longrightarrow}} C(f) \longrightarrow X[1] \longrightarrow \tag{4}$$

[Crossed out above the arrow $Y \to C(f)$: illegible entry$^?$.]



Uncertain reading. illegible entry

Original source details: GELFAND-P001397

[Crossed out before $C(f)$: X? / Y?$^?$.]

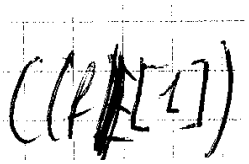


Uncertain reading. X? / Y?

Original source details: GELFAND-P001397

[Below $X[1]$: the grading is shifted and the sign is changed on d.]

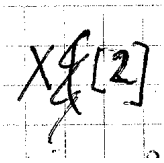
$$\longrightarrow Y(1) \longrightarrow C(f[1]) \longrightarrow X[2] \longrightarrow$$



Original source details: GELFAND-P001397

Uncertain reading. The closing parenthesis follows the shift. The expression reads $C(f[1])$.

[Between X and $[2]$ crossed out: a mark resembling a parenthesis?]



Original source details: GELFAND-P001397

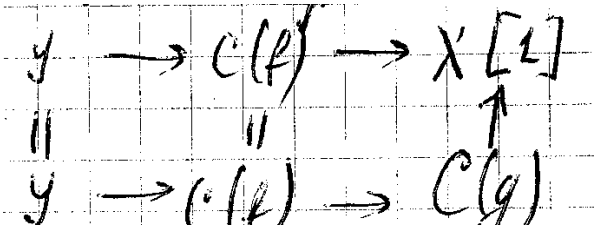
Uncertain reading. a mark resembling a parenthesis

it goes into itself under a shift of three steps. In the homotopy category it goes into itself under a shift of one step.

$$Y \xrightarrow{g} C(f) \longrightarrow C(g) \tag{5}$$

(6) Assertion. In $K(A)$ the complex $C(g) = X[1]$.

There is a map:



Original source details: GELFAND-P001397

Diagram and typed description.

$$\begin{array}{ccccc} Y & \longrightarrow & C(f) & \longrightarrow & X[1] \\ \| & & \| & & \uparrow \\ Y & \longrightarrow & C(f) & \longrightarrow & C(g) \end{array}$$

Namely,

$$C(g)^i = C(f)^i \oplus Y^{i+1} = X^{i+1} \oplus Y^i \oplus Y^{i+1}$$

projects to X^{i+1}.

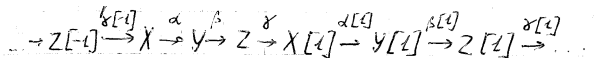
GELFAND-P001398

[source number: 24]

Exercise 2. Check that this is a morphism of complexes and an isomorphism in the homotopy category.

Exercise 3. Find in $K(A)$ the cones $C(\underset{\text{identity map}}{\underset{|}{id}})$ and $C(\underset{\text{zero map}}{\underset{|}{0}})$.

Definition 1. A triangle in $K(A)$ or $D(A)$ is the diagram



Original source details: GELFAND-P001398

Diagram and typed description.

$$\cdots \longrightarrow Z[-1] \xrightarrow{\gamma[-1]} X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \xrightarrow{\gamma} X[1] \xrightarrow{\alpha[1]} Y[1] \xrightarrow{\beta[1]} Z[1] \xrightarrow{\gamma[1]} \cdots$$



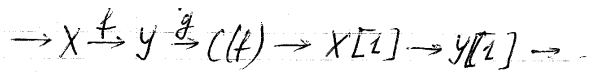
Original source details: GELFAND-P001398

Diagram and typed description.



Crossed-out sketch. Two arrows are visible; any labels are illegible.

Def. 2. An exact triangle in $K(A)$ or $D(A)$ is a triangle isomorphic to the triangle



Original source details: GELFAND-P001398

Diagram and typed description.

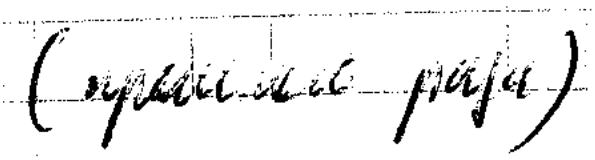
$\longrightarrow X \xrightarrow{f} Y \xrightarrow{g} C(f) \longrightarrow X[1] \longrightarrow Y[1] \longrightarrow \cdots$

GELFAND-P001399

9/XI-87 Gelfand seminar

[Crossed out: Lascoux.] Lascoux

I. M. G. after Lascoux (beautiful / regular[?] series)



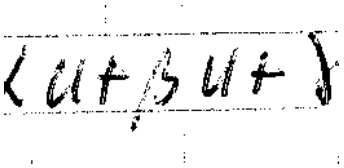
Original source details: GELFAND-P001399

Uncertain reading. beautiful / regular

Schubert polynomials.

Consider the polynomial

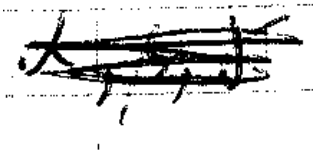
$$P = \alpha u + \beta u + \gamma w + \cdots \tag{1}$$



Original source details: GELFAND-P001399

Uncertain reading. Second monomial in (1): appears to read βu, possibly βv.

u, v, w [Crossed out: possibly α, β, γ[?]] are monomials.



Original source details: GELFAND-P001399

Uncertain reading. possibly α, β, γ

In our case we will have $u, v, w,$ [Crossed out: illegible mark] $\ldots = x_1, x_2, \ldots, x_n$.



Original source details: GELFAND-P001399

Uncertain reading. illegible mark

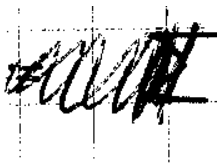
$\alpha, \beta, \gamma, \ldots$ are numbers.

The polynomials [Crossed out: fn. / fns.?] $S^m(P)$ [Crossed out: possibly Schur?] are defined by the formula



Original source details: GELFAND-P001399

Uncertain reading. fn. / fns.



Original source details: GELFAND-P001399

Uncertain reading. possibly Schur

$$\sum_{m\geq 0}^{\infty} z^m S^m(P) = \frac{1}{(1-zu)^\alpha (1-zv)^\beta \ldots} \tag{2}$$



Original source details: GELFAND-P001399

Uncertain reading. Lower limit in (2): reads as $m \geq 0$.

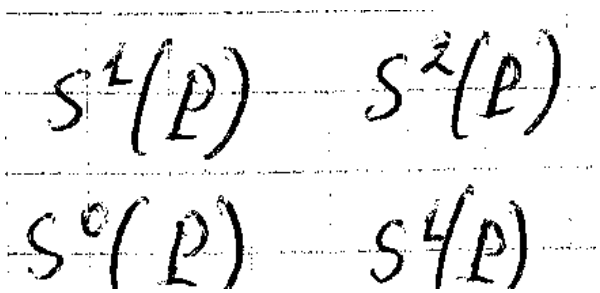
$$\alpha = \beta = \gamma = \cdots = 1 \quad \Rightarrow$$
$$S^m(x_1 + x_2 + \cdots + x_n) = \sum_{i_1 \leq i_2 \leq \cdots \leq i_m} x_{i_1} x_{i_2} \cdots x_{i_m} \tag{3}$$

For example,

$$S^2(x_1 + x_2) = x_1^2 + x_1 x_2 + x_2^2.$$

Consider a Hankel matrix (of infinite order)

$$\begin{pmatrix} S^0(P) & S^1(P) & S^2(P) & \ldots \\ S^{-1}(P) & S^0(P) & S^1(P) & \ldots \\ & & & \end{pmatrix}, \qquad S^{-i}(P) = 0 \text{ for } i > 0. \tag{4}$$



Original source details: GELFAND-P001399

Uncertain reading. Exponent in position $(2,3)$ of (4): more likely 1 than 4.

(see Akhiezer's book on the moment problem)

Multiplication of the series (2) corresponds to multiplication of the matrices (4).

GELFAND-P001400

[source number: 30]

A topic in operator theory that I. M. G. suggested to Birman / Burman[?]:

Uncertain reading. Birman / Burman



Original source details: GELFAND-P001400

Construct / relate[?] K-functor in operator theory

Uncertain reading. Construct / relate



Original source details: GELFAND-P001400

(Sato did this 15 years later)

and it came into fashion.

The only theorem from matrix theory

Consider the matrix A: $f_{\alpha\beta} = f(\alpha, \beta)$ and all its minors of order k: this is also a matrix (the associated matrix of order k) $\Lambda^k(A)$. Then

$$\Lambda^k(AB) = \Lambda^k(A)\Lambda^k(B) \tag{5}$$

It is convenient to denote the row indices by $i_1 + 1, i_2 + 2, \ldots, i_p + p$, and the column indices by $j_1 + 1, j_2 + 2, \ldots, j_p + p$

$$\begin{vmatrix} i_1 \leq i_2 \leq \cdots \leq i_p \\ j_1 \leq \cdots \leq j_p \end{vmatrix}$$

$$S_{J|I}(P) \tag{6}$$

is the corresponding minor, called the skew Schur function.

$S_{J|0}(P)$ is the ordinary Schur function.

[Crossed out: Then] One can consider the polynomial (Jacobi)

$$\frac{\begin{vmatrix} x_1^{j_1} & x_1^{j_2+1} & \ldots & x_1^{j_p+p-1} \\ x_2^{j_1} & x_2^{j_2+1} & \ldots & x_2^{j_p+p-1} \\ \ldots & \ldots & \ldots & \ldots \end{vmatrix}}{\begin{vmatrix} 1 & x_1 & \ldots & x_1^{p-1} \\ 1 & x_2 & \ldots & x_2^{p-1} \\ & & & \\ 1 & x_p & \ldots & x_p^{p-1} \end{vmatrix}} \tag{7}$$



Uncertain reading. In the first two rows of the numerator, probably $+1$ is crossed out after j_1.

Original source details: GELFAND-P001400



Uncertain reading. To the right of the numerator: 1 or a single oblique stroke.

Original source details: GELFAND-P001400

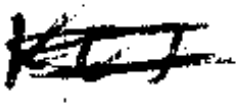
(8 / $7^?$) ([Inserted above the line: Ex.] check that the denominator divides the numerator!)



Uncertain reading. 8 / 7

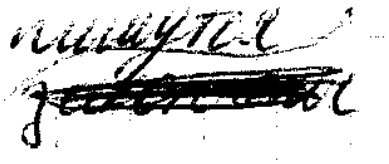
Original source details: GELFAND-P001400

[Crossed out: ко / но [but]?.] This is a symmetric function, but the coefficients are not [Crossed out: illegible] [Inserted above the line: written down explicitly] and are still being studied.



Uncertain reading. ко / но [but]

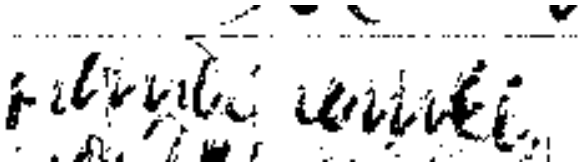
Original source details: GELFAND-P001400



Undeciphered passage. word under the inserted «are written down» is illegible

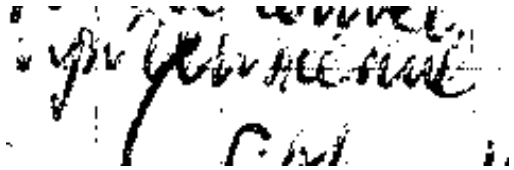
Original source details: GELFAND-P001400

[In the left margin: (first steps,? possibly «in solving it»?)]



Uncertain reading. first steps,

Original source details: GELFAND-P001400



Uncertain reading. possibly «in solving it»

Original source details: GELFAND-P001400

(see the paper by [crossed-out initials: I. G. / I. M.?; Inserted above the line: I. M.] Gelfand, Zelevinsky, 1986, FA; and the unpublished work by B. Lidskii / B. Litskii?)



Uncertain reading. I. G. / I. M.

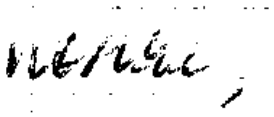
Original source details: GELFAND-P001400



Uncertain reading. B. Lidskii / B. Litskii

Original source details: GELFAND-P001400

It is clear so far / only?, that the coefficients [Crossed out: will be / all?] are integers.



Uncertain reading. so far / only

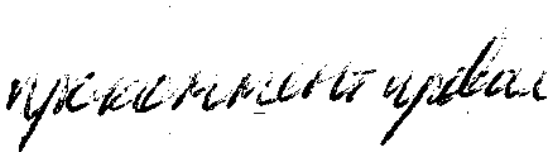
Original source details: GELFAND-P001400



Original source details: GELFAND-P001400

Uncertain reading. will be / all

Schur realized that they are important: he commented on / having commented on[?] Jacobi, Frobenius and others.



Original source details: GELFAND-P001400

Uncertain reading. he commented on / having commented on

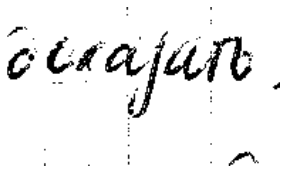
GELFAND-P001401

Frobenius found the characters of the symmetric group. This is a deep work for which H. Weyl's commentaries are no substitute.

(9) Theorem. (Jacobi–Trudi)

$$(7) = S_{J|0}(x_1 + \ldots + x_p) \tag{10}$$

(11) Problem: a) possibly «grasp it» (осознать)[?];



Original source details: GELFAND-P001401

Uncertain reading. possibly «grasp it» (осознать)

b) give its classical proof;

c) give an interpretation in representation theory.

(12) $S_{J|I}$ satisfies a system of identities;

these are Plücker coordinates (one of Sato's achievements!)

$$S^m(P+Q) = \sum_j S^{n-j}(P)S^j(Q); \tag{13}$$



Original source details: GELFAND-P001401

Uncertain reading. The exponent on the left of (13) is overwritten and looks like m or n.

$$S^n(PQ) = \sum_{|J|=n} S_J(\underline{P})S_J(Q), \qquad S'_J = S'_{J|0}; \tag{14}$$

$$S^n(S^m(P)) = \sum (n, m; J)S_J(P) \quad \|, \tag{15}$$



Uncertain reading. Two strokes resembling $\|$ follow $S_J(P)$ in (15).

Original source details: GELFAND-P001401

where $(n, m; J)$ are universal coefficients.

The cohomology of flag manifolds is computed in two ways

a) decomposition into Schubert cells (the dimensions are even!)

b) differential forms;

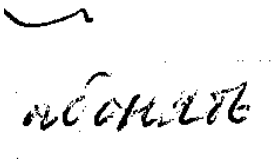
Which forms correspond to which cells?

Schubert polynomials arise there

(up to symmetric ones)

Lascoux, Schützenberger: one should forget topology and compute polynomials.

The polynomials should be housed[?] in the ring of all polynomials, and so on.



Uncertain reading. housed

Original source details: GELFAND-P001401

GELFAND-P001402

[source number: 31]

[The following passage is boxed and crossed out.]

(16) Theorem. Let us permute the rows

$$h_1 + 1,\ h_2 + 2, \ldots, h_p + p, \qquad h_1 \leq h_2 \leq \ldots \leq h_p;$$

Newton introduced the operations

$$\partial_i f(x_1, \ldots, x_n) = \frac{f(x_1, \ldots, x_{i+1}, x_i, \ldots, x_n) - f(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n)}{x_{i+1} - x_i} \tag{16}$$

Then

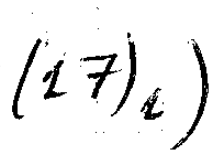
$$\begin{cases} \partial_i^2 = 0 \\ \partial_i \partial_j = \partial_j \partial_i, \qquad |i - j| \geq 2, \\ \partial_i \partial_{i+1} \partial_i = \partial_{i+1} \partial_i \partial_{i+1} \end{cases} \tag{17}$$

(18) Remark.

$(\partial_i + 1)(\partial_i - q) = 0 \quad \Rightarrow \quad \partial_i$ is a q-analogue of the generators of the symmetric group

$q = 1 \quad \Rightarrow \quad$ symmetric group

(instead of $(17)_1$)



Uncertain reading. The subscript on (17) reads as 1.

Original source details: GELFAND-P001402

Lascoux

Connection between Schubert polynomials and Schur functions.

(19) $\partial_i(fg) = g\partial_i(f)$, if [crossed out: appears to be f?] g is symmetric in [crossed out: two / all?] the variables x_i, x_{i+1}.



Uncertain reading. appears to be f

Original source details: GELFAND-P001402



Uncertain reading. two / all

Original source details: GELFAND-P001402

Reminder about Schubert polynomials

$$Y_{730} = x_1^7 x_2^3 x_3^0$$

(20) Def. If $\underbrace{i_1 \geq i_2 \geq i_3 \geq \ldots}_{I}$, then $x_1^{i_1} x_2^{i_2} \ldots = Y_I$

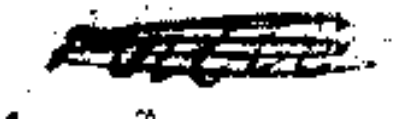
(Schubert polynomial [crossed out: I / $=^?$] [crossed out: illegible] with index I is a monomial)

Uncertain reading. I / $=$



Original source details: GELFAND-P001402

Undeciphered passage. crossed-out writing is illegible



Original source details: GELFAND-P001402

All Schubert polynomials $=$ are the images of these under the action of an arbitrary operator ∂_i.

Uncertain reading. appears to be $=$



Original source details: GELFAND-P001402

[Inserted above «images»: «(all nonzero)».]

Rule for calculating$^?$ the indices

Uncertain reading. calculating



Original source details: GELFAND-P001402

$$Y_{\ldots i_p\, i_{p+1}\ldots} \xrightarrow{\partial_p} \begin{cases} Y_{\ldots i_{p+1}\, i_p - 1}, & i_p > i_{p+1} \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

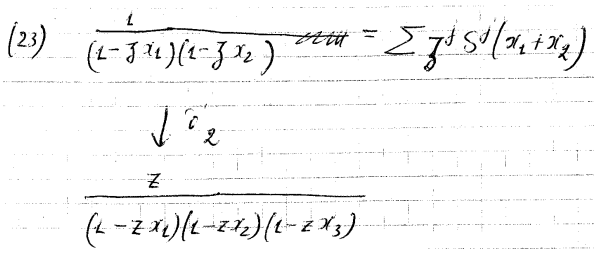
Ex. Prove that these definitions are well defined.

(22) Example

$$Y_{730} = x_1^7 x_2^3 x_3^0 = \begin{vmatrix} S^3(x_1 + x_2) & S^8(x_1) \\ S^2(x_1 + x_2) & S^7(x_1) \end{vmatrix}$$

GELFAND-P001403

(23)



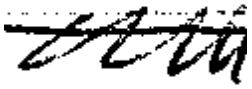
Original source details: GELFAND-P001403

Diagram and typed description.

$$\frac{1}{(1-zx_1)(1-zx_2)} = \sum z^j S^j(x_1+x_2)$$

$$\downarrow \partial_2$$

$$\frac{z}{(1-zx_1)(1-zx_2)(1-zx_3)}$$



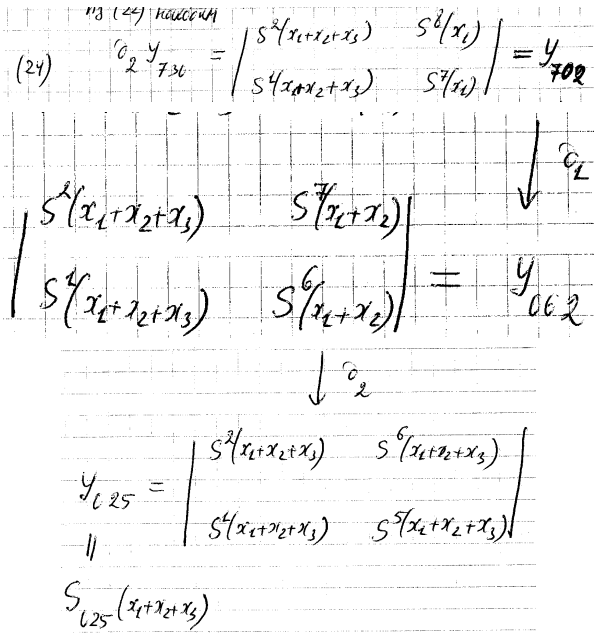
Original source details: GELFAND-P001403

Undeciphered passage. The crossed-out writing after the fraction, before the equals sign, is illegible.

$$S^n(x_1+x_2) \xrightarrow{\partial_2} S^{n-1}(x_1+x_2+x_3) \tag{23}$$

From (22) we obtain

(24)



Original source details: GELFAND-P001403

Diagram and typed description.

$$\partial_2 Y_{730} = \begin{vmatrix} S^2(x_1+x_2+x_3) & S^8(x_1) \\ S^1(x_1+x_2+x_3) & S^7(x_1) \end{vmatrix} = Y_{702}$$

$$\downarrow \partial_1$$

$$\begin{vmatrix} S^2(x_1+x_2+x_3) & S^7(x_1+x_2) \\ S^1(x_1+x_2+x_3) & S^6(x_1+x_2) \end{vmatrix} = Y_{062}$$

$$\downarrow \partial_2$$

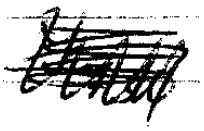
$$Y_{025} = \begin{vmatrix} S^2(x_1+x_2+x_3) & S^6(x_1+x_2+x_3) \\ S^1(x_1+x_2+x_3) & S^5(x_1+x_2+x_3) \end{vmatrix}$$

$$\| $$

$$S_{025}(x_1+x_2+x_3)$$

(Schur function, defined earlier; $S_{025|0}(x_1+x_2+x_3)$, sequences of zeros are omitted)

The expression of formula [crossed out: illegible] (22) as a determinant was used in elimination theory in the 19th century.



Original source details: GELFAND-P001403

Undeciphered passage. word illegible, possibly «Schur»

There are 5 (thick) volumes of the book

T. Muir: History of Determinants, Dover (1960)

GELFAND-P001404

[source number: 32]

There is another book (by the same author)

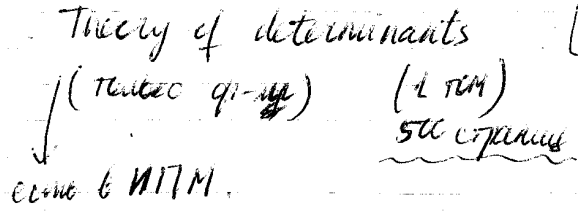
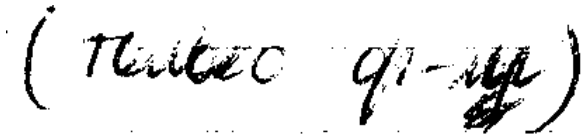


Diagram and typed description.
Theory of determinants
(formulas only) (1 volume)
↓ 500 pages
available at the IPM.

Original source details: GELFAND-P001404



Uncertain reading. formulas only / thousands of formulas

Original source details: GELFAND-P001404

[To the right of the title] everything up to 1900 | almost everything up to (1920)

Analogy: theory of integrable systems, construct / build[?] a theory to account for these formulas.

Uncertain reading. construct / build

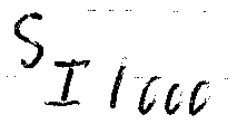


Original source details: GELFAND-P001404

(25) Theorem.

$$S_{I|000}(x_1 + \cdots + x_n) = Y_{i_1 \ldots i_n 0000}$$

Uncertain reading. Index on the left: $I \mid 000$.



Original source details: GELFAND-P001404

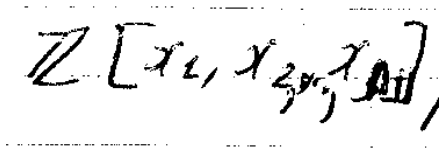
$$0 \leq i_1 \leq \cdots \leq i_n$$

(26) Finite case: $\mathbb{Z}[x_1, x_2, \ldots, x_{n+1}]$ / ideal of symm. pol. in $(x_1, x_2,$ [crossed out: an x is visible, with an illegible subscript] $\ldots, x_{n+1})$

Uncertain reading. an x is visible, with an illegible subscript



Original source details: GELFAND-P001404



Original source details: GELFAND-P001404

Uncertain reading. The final subscript before the bracket reads as $n+1$.

(27) Theorem. The Schubert polynomials

$$Y_I, \quad 0 \le i_1 \le n, \quad 0 \le i_2 \le n-1, \quad \dots, \quad 0 \le i_{n+1} \le 0 \tag{28}$$

form a basis of the ring (26) as a free abelian group.

Example

$n = 2$ $\mathfrak{S}(3)$;

$$Y_{210} \quad Y_{110} \quad Y_{200} \quad Y_{010} \quad Y_{100} \quad Y_{000}$$

$\mathbb{Z}$-basis in $\mathbb{Z}[x_1, x_2, x_3]/(\quad)$

From (27) it is clear that the dimension is $n!$.

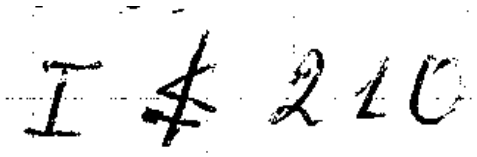


Original source details: GELFAND-P001404

Uncertain reading. The manuscript reads $n!$.

The others lie in the ideal:

$$Y_I \in \text{ideal} \quad \text{if} \quad I \not\le 210 \tag{29}$$



Original source details: GELFAND-P001404

Uncertain reading. The sign in (29) is a struck-through $\le$.

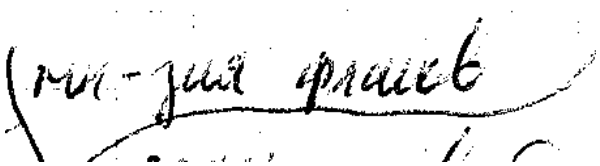
(30) Question of I. M. G.: how is the representation of $\mathfrak{S}(n)$ [crossed out: possibly «в» (in)?] written when $\mathfrak{S}(n)$ acts on the variables $x_1, \dots, x_n$ in the Schubert polynomials



Original source details: GELFAND-P001404

Uncertain reading. possibly «в» (in)

In the cohomology ring [inserted above the line with an insertion mark: of the flag manifold?] there is a quadratic form, written as follows:



Original source details: GELFAND-P001404

Uncertain reading. of the flag manifold

$$PQ \xrightarrow{\partial_1 \partial_2 \partial_1} (P, Q) \tag{31}$$

(scalar product)

[To the right:] (a number in the quotient by the ideal)

(32) Chevalley theorem.

∂_1 is self-adjoint with respect to $(\ ,\ )$

∂_2 —"—

e.g.

$$(\partial_1(P), Q) = (P, \partial_1(Q)) \tag{33}$$

The proof is straightforward.

Lascoux.

$$Y_I, \qquad Y_{I'}, \qquad I + I' = n \quad n-1 \quad \dots \quad 1 \quad 0 \tag{34}$$

GELFAND-P001405

Then

$$(Y_I, Y_J) = 0 \quad \text{when } J \neq I'. \tag{35}$$

Uncertain reading. The symbol in (35) reads as 0.



Original source details: GELFAND-P001405

(all in the quotients)

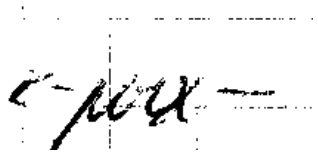
Theorem.

(36) $\mathbb{Z}[x_1, x_2, x_3]$ is a free module [inserted above the line: over] Sym(x_1, x_2, x_3) with the Schubert polynomials as a basis.

Not all Schubert polynomials are determinants, so divided differences are used in their place, ∂_i.

Problem of I. M. G. Generalize the theorem on characters: irreducible representations, characters of the

Uncertain reading. «с-ких» [uncertain abbreviation]



Original source details: GELFAND-P001405

— Schubert functions.

Next time, Takhtajan on quantum groups.

Kapranov (Misha)

2-categories.

This is an algebraic structure with objects of 4 types

a) objects •

b) horizontal morphisms • ⟶ •

c) vertical morphisms ↑

d) squares

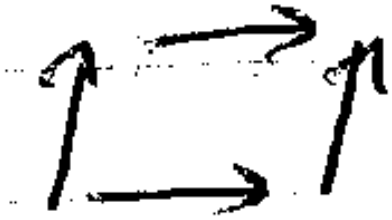


Diagram and typed description. 

Original source details: GELFAND-P001405

[In the right margin:]

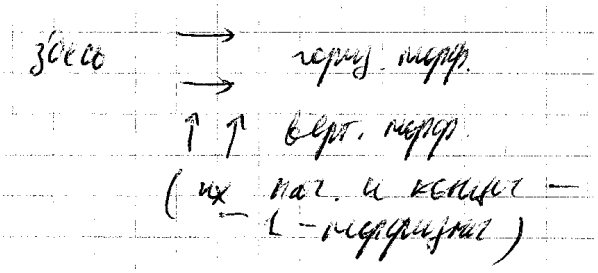


Diagram and typed description. Here ⟶ ⟶ horizontal morphisms,
↑↑ vertical morphisms
(their sources and targets are 1-morphisms)

Original source details: GELFAND-P001405

One can compose:

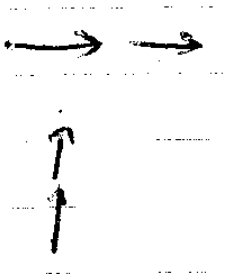


Diagram and typed description.
1) ⟶ ⟶ 2 horizontal morphisms
2) ↑ ↑ 2 vertical morphisms

Original source details: GELFAND-P001405

3) squares are multiplied in two ways

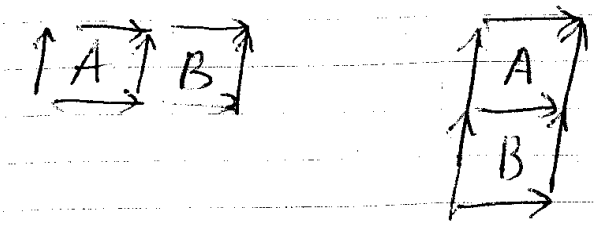
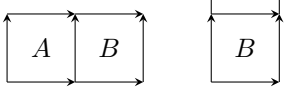
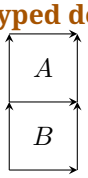


Diagram and typed description.





Original source details: GELFAND-P001405

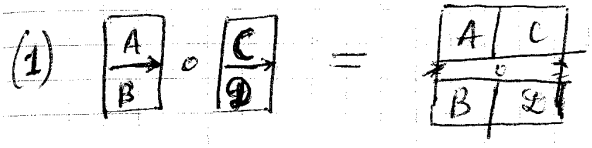
(see Gabriel's book)

GELFAND-P001406

[source number: 33]

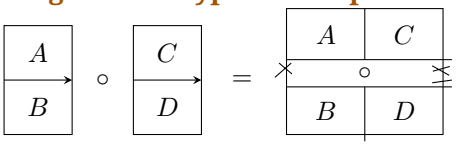
Main nontrivial axiom: 2-dimensional associativity

(1)



Original source details: GELFAND-P001406

Diagram and typed description.



The reading D is tentative. The strokes at the ends of the band on the right leave the arrows uncertain.

Example where [crossed out: illegible] ax. fails [crossed out: possibly an earlier «(1)»[?]] (1)



Original source details: GELFAND-P001406

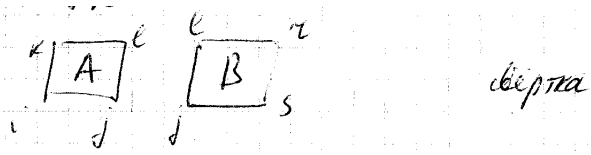
Undeciphered passage. illegible



Original source details: GELFAND-P001406

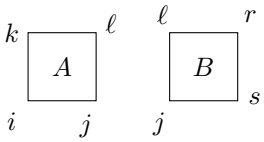
Uncertain reading. possibly an earlier «(1)»

Tensors:



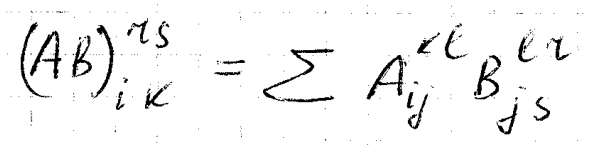
Original source details: GELFAND-P001406

Diagram and typed description.



contraction

$$(AB)^{rs}_{ik} = \sum A^{k\ell}_{ij} B^{\ell r}_{js}$$



Original source details: GELFAND-P001406

Uncertain reading. On the left, i, k are subscripts and r, s are superscripts.

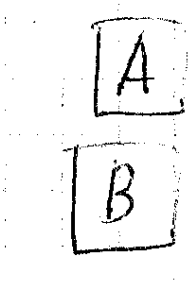


Diagram and typed description. A B

Original source details: GELFAND-P001406

Axiom (1) will fail.

Another def. (M. Kontsevich) (competing)

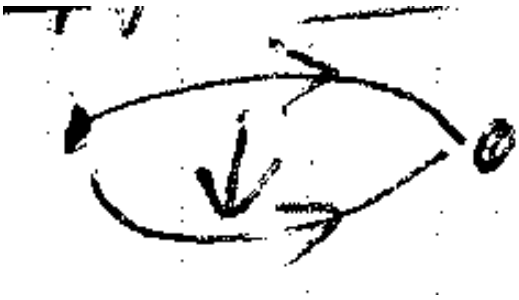
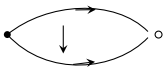


Diagram and typed description. 
morphisms between morphisms with the same endpoints.

Original source details: GELFAND-P001406

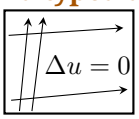
Example: Cat (the category of categories. There is a morphism of functors)

This is a special case of the preceding one (contract the vertical arrows).

I. M. G.



Diagram and typed description.

св.? u and $\frac{\partial u}{\partial n}$ 

boundary conditions

The abbreviation above u is unclear.

Original source details: GELFAND-P001406

Request to Keldysh, Chentsov, and Gelfand

K. Higher fundamental groups. $X_0 \subset X_1 \subset X$ is a topological space

objects are points $x \in X_0$

morph. $\left(\underset{\text{interval}}{I}, \partial I\right) \longrightarrow (X_1, X_0)$ homotopic $(\text{mod } \partial I)$ are regarded as equivalent.

bimorphisms $(I^2, \partial I^2) \longrightarrow (X, X_1)$ (mod homotopy)

GELFAND-P001407

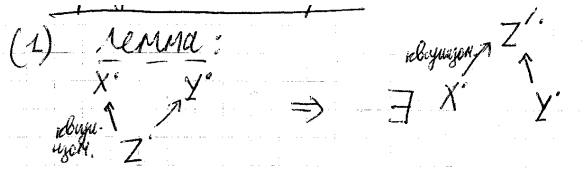
[source number: 44]

23/XI-87 Gelfand seminar.

Beilinson (continued)

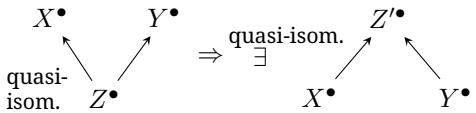
Derived categories.

(1) Lemma:



Original source details: GELFAND-P001407

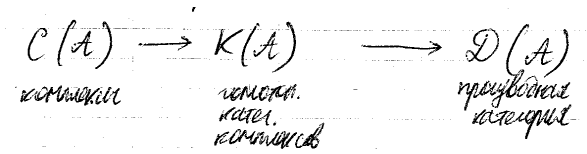
Diagram and typed description.



The lemma is proved using the mapping cone construction.

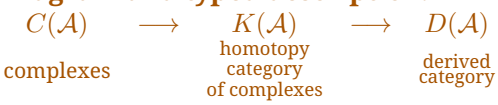
Reminder:

$\mathcal{A}$ is an ab. category



Original source details: GELFAND-P001407

Diagram and typed description.



[At right:] (the objects are complexes throughout)

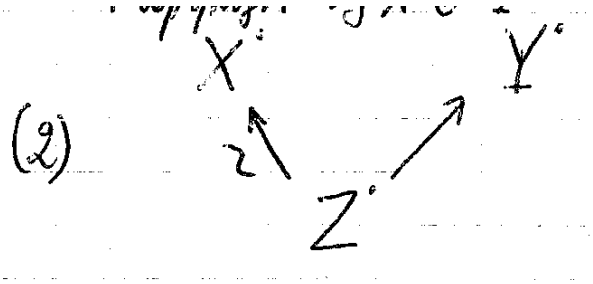
$f : X^\bullet \to Y^\bullet$ is homotopic to 0 if $\exists g^i : X^i \to Y^{i-1}$,

$$f^i = dg^i + g^{i+1}d.$$

$D(\mathcal{A})$ is $K(\mathcal{A})$ with the following inverted quasi-isomorphisms (morphisms inducing an isomorphism of cohomology)

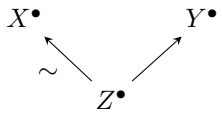
A morphism from $X^\bullet$ to $Y^\bullet$ is a diagram in $K(\mathcal{A})$

(2)



Original source details: GELFAND-P001407

Diagram and typed description.



(the left arrow is a quasi-isomorphism)

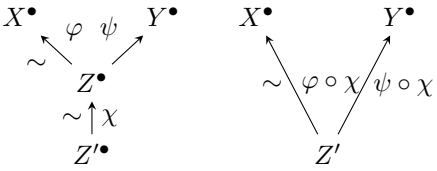
more precisely, an equivalence class of such diagrams. Diagram (2) is equivalent to the diagram obtained using any quasi-isomorphism $\chi : Z'^{\bullet} \to Z^{\bullet}$

(3)



Original source details: GELFAND-P001407

Diagram and typed description.



(this relation extends to the smallest equivalence relation containing it)

The cone of a morphism and exact triangles were defined [crossed out: illegible] already.

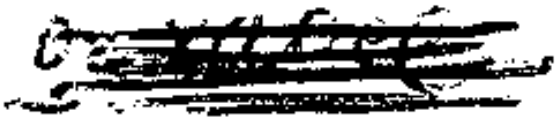


Uncertain reading. The short crossed-out passage before «already» is illegible.

Original source details: GELFAND-P001407

GELFAND-P001408

I. Zakharevich [crossed out: illegible] Improvisation on chaos.



Undeciphered passage. illegible

Original source details: GELFAND-P001408



Uncertain reading. The preposition in the title reads as «o» (on).

Original source details: GELFAND-P001408

Examples

1°. Apparently there is a morphism from one example to the other.

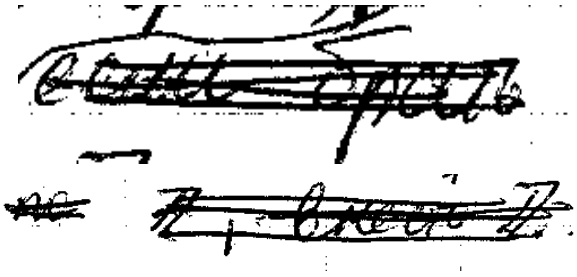
$$M = (\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}} \qquad \text{(sequences infinite in both directions)}$$

$T : M \to M$. M has the standard topology (a Cantor set).

$$a_i \to a_{i-1} \qquad \text{(shift transformation, Bernoulli transformation)}$$

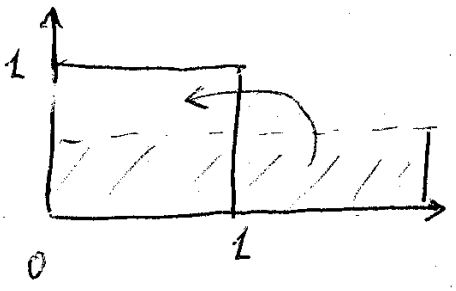
Consider the iterates T^N.

There is a measure on M. The baker's transformation coincides with this [inserted above the line: (up to a set of measure 0)] [crossed out: continuation on two lines].



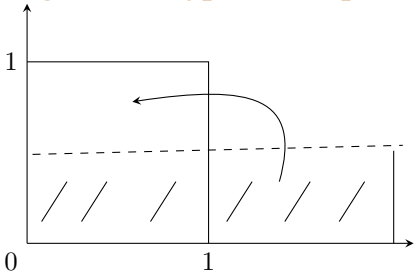
Original source details: GELFAND-P001408

Uncertain reading. The upper line appears to read «if one takes», the lower «not $\mathbb{Z}$, instead of $\mathbb{Z}$».



Original source details: GELFAND-P001408

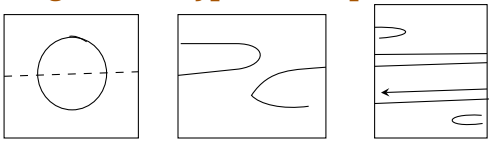
Diagram and typed description.





Original source details: GELFAND-P001408

Diagram and typed description.



and so on.

The small strokes on the curves in the first and third drawings leave their direction uncertain.

Correspondence: write the coordinates of a point of the square in binary and take one digit to the left, the other to the right.

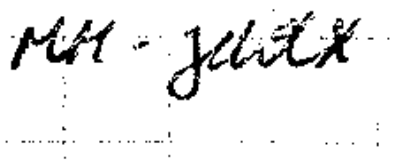
In what sense is this chaos

Consider $T^k x,\ k \leq 0$ — suppose this is known (to within 1 mm).

$$T^k x, \qquad k > 0$$

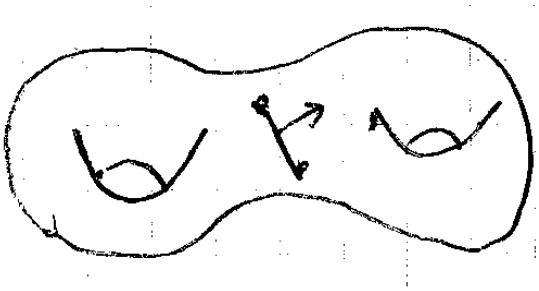
In a finite time all this information will be lost.

2°. Hyperbolic transformation: geodesic flows on manifolds of negative curvature.



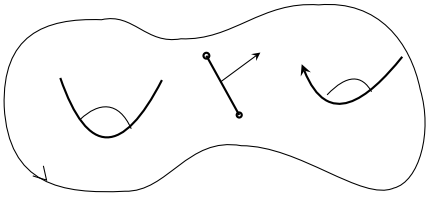
Original source details: GELFAND-P001408

Uncertain reading. The abbreviation reads as «мн-зиях» (manifolds).



Original source details: GELFAND-P001408

Diagram and typed description.



The middle branch points up and right. The stroke near the left edge of the outline and the small marks on the middle segment are uncertain.

GELFAND-P001409

[source number: 45]

G. Margulis. Iterations of maps of the complex plane.

Consider the transformation $z \to P(z)$, P is a polynomial

(more generally, a rational function)

$\deg P = 1$ [crossed out: illegible] uninteresting case



Original source details: GELFAND-P001409

Undeciphered passage. The short passage after $\deg P = 1$ is illegible.

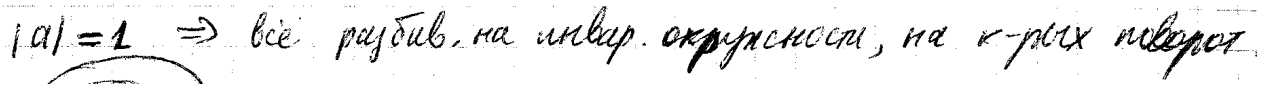
$$z \mapsto az + b \tag{1}$$

Iterations:

$$z_{i+1} = P(z_i)$$

$|a| > 1 \Rightarrow$ one fixed point.

$|a| = 1 \Rightarrow$ everything splits into invariant circles on which the map is a rotation.



Original source details: GELFAND-P001409

Uncertain reading. The manuscript reads $|a| = 1$.

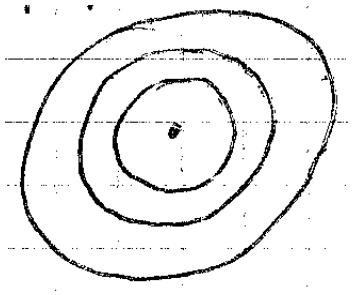
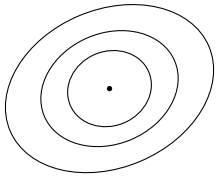


Diagram and typed description.



Original source details: GELFAND-P001409

I. M. G.: 1) the question of when a transformation of the circle is diffeomorphically conjugate to a rotation is nontrivial. Denjoy, Arnold, Herman.

2) What is a representation of an infinite discrete group?

There is no satisfactory answer.

Von Neumann's theorem is unaesthetic.

Chaos arises on the circle.

Conjecture: if one works with sets of intermediate Hausdorff dimension, some of the classical theory will be recovered!!

Question for school students: cover the nth rational number with an interval of length $\varepsilon/2^n$. Give an example of a number that will remain uncovered.

Problem: find the Hausdorff dimension of the perfect Cantor set.

(!) Problem. Construct a theory of representations of discrete groups on functions on sets of intermediate Hausdorff dimension.

To begin, look at physics, string theory and so on.

G. M.: $z \to z^2$ $\quad |z| = 1$ maps to itself

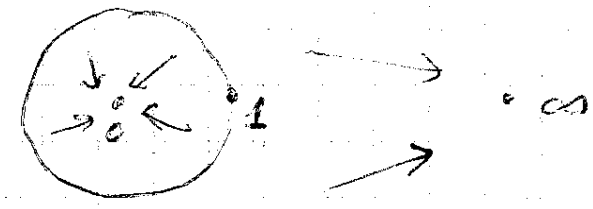
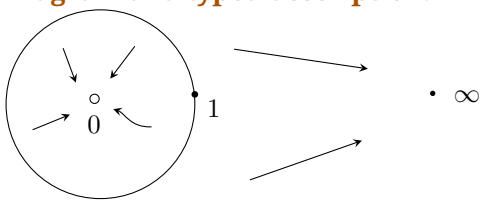


Diagram and typed description.



Original source details: GELFAND-P001409

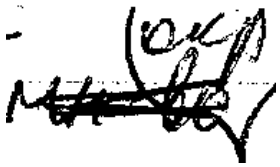
on the circle, an analogue of the (one-sided) baker's transformation

GELFAND-P001410

$$f_n = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}$$

Def.

U is a [crossed out: set; inserted above the line: nbhd.] of normality if $f_n|_U$ is a compact set (normal family of analytic functions)



Original source details: GELFAND-P001410

Uncertain reading. «set» is crossed out and «nbhd.» is inserted above it.

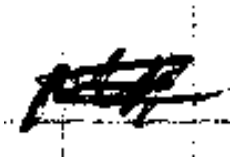
[Crossed out on a short line by itself:]



Original source details: GELFAND-P001410

Undeciphered passage. illegible

Set of normality [inserted above the line: X] consists of points x, for which $\exists$ nbhd. U — [crossed out: illegible] a nbhd. of normality



Original source details: GELFAND-P001410

Undeciphered passage. The crossed-out writing before «nbhd. of normality» is illegible.

[Under Set of normality: (it is open)]

The complement of the set of normality is the Julia set.

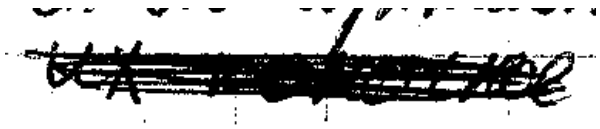
(it is closed)

The set of normality can be empty

$$X = X_1 \cup \cdots \cup X_n \cup \cdots$$

(connected components) of the set of normality

[crossed out below this caption: illegible]



Original source details: GELFAND-P001410

Undeciphered passage. Individual letters are visible below the caption, but the passage cannot be read as a whole.

$$f : (\text{connected comp.}) \to (\text{connected components})$$
$$X_i \mapsto X_{j(i)}$$

Sullivan proved that for the resulting [crossed out: illegible] map of a discrete set (countable or finite)



Undeciphered passage. The crossed-out writing before «map» is illegible.

Original source details: GELFAND-P001410

a) each orbit stabilizes,

b) the set of fixed points is finite.

Idea of the proof. Suppose there is an infinite orbit

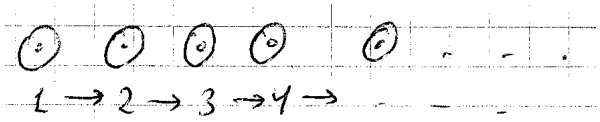
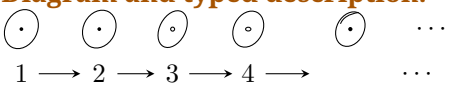
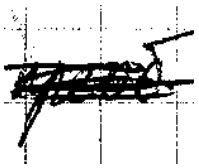


Diagram and typed description.



Original source details: GELFAND-P001410

Let us change the conformal [crossed out: possibly «transf.»] structure so that the transformation remains conformal.



Uncertain reading. The crossed-out word appears to read «transf.».

Original source details: GELFAND-P001410

Problem:

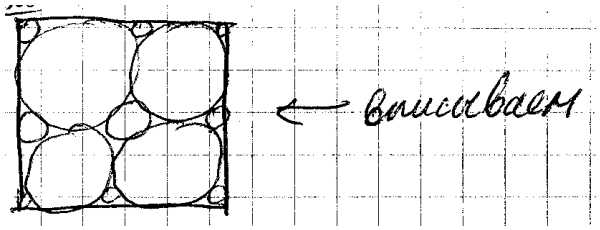
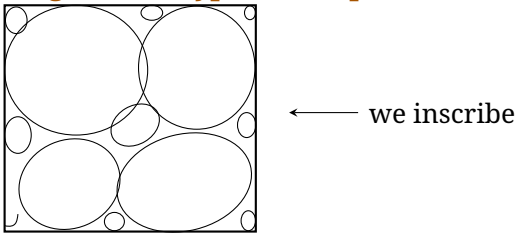


Diagram and typed description.



The contacts and some small arcs at the boundary are shown approximately.

Original source details: GELFAND-P001410

1) Prove that the remainder has measure 0

2) Prove that the remainder has Hausdorff dimension < 2.

GELFAND-P001411

[source number: 46]

I. M. Gelfand. Matroids.

In a K-dimensional space, consider n vectors. Number them and list which are linearly dependent and which are not.

Giving this list defines a matroid.

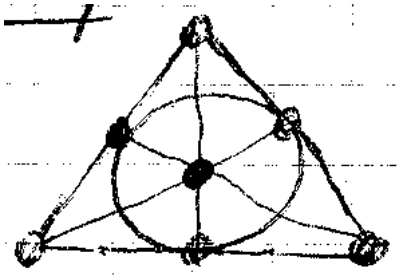
What conditions apply to this list?

Example: [Crossed out: illegible]



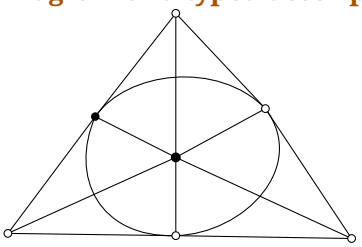
Original source details: GELFAND-P001411

Undeciphered passage. The writing after «Example» is illegible.



Original source details: GELFAND-P001411

Diagram and typed description.



If 3 points lie on one line, the vectors are linearly dependent.

The whole space is 3-dimensional (no 4 linearly independent vectors)

(1) Problem: 1) prove that over $\mathbb{R}$ and $\mathbb{C}$ there are no such vectors!

2) over $\mathbb{Z}/2\mathbb{Z}$ there are such vectors.

(all the nonzero vectors of $(\mathbb{Z}/2\mathbb{Z})^3$!)

General (unsolved problem): which matroids are realizable a) over $\mathbb{R}$ b) over $\mathbb{C}$.?

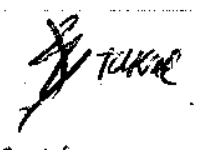
Solved problem: which matroids are realizable over $\forall$ fields?

In Rota's book э...[At right: see book? by Aigner (?)] there are 15 different equivalent definitions of a matroid.



Original source details: GELFAND-P001411

Uncertain reading. After «Rota» there is a mark resembling «э» followed by dots.



Original source details: GELFAND-P001411

Uncertain reading. The word after «see» has been overwritten and tentatively reads «книге» or «книжке» (book).

(2) Axioms.

Certain subsets are singled out, called bases

B_1, B_2 are 2 bases

$$y \in B_2,\ y \notin B_1 \Rightarrow \exists x \in B_1; \qquad (B_1 \setminus x) \cup y, \quad (B_2 \setminus y) \cup x$$

are bases [Crossed out: writing between the two expressions]



Original source details: GELFAND-P001411

Undeciphered passage. The writing after $(B_1 \setminus x) \cup y$ is crossed out and illegible.

Exercise in linear algebra: verify this for a set of vectors in a linear space.

(3) Bimatroid: consider a matrix. It defines 2 matroids, one by columns and one by rows S and T

Another notion: list all the minors $\neq 0$

Bimatroid: a collection of subsets with the same number of elements.

(4) [Crossed out: illegible] Problem: give axioms for a bimatroid.



Original source details: GELFAND-P001411

Undeciphered passage. The short passage before «Problem» is illegible.

GELFAND-P001412

(5) Problem: how is the notion of a bimatroid from [Crossed out: illegible] Gelfand–Serganova (UMN) related to «Кунтом»



Original source details: GELFAND-P001412

Undeciphered passage. The crossed-out writing after “from” is illegible.

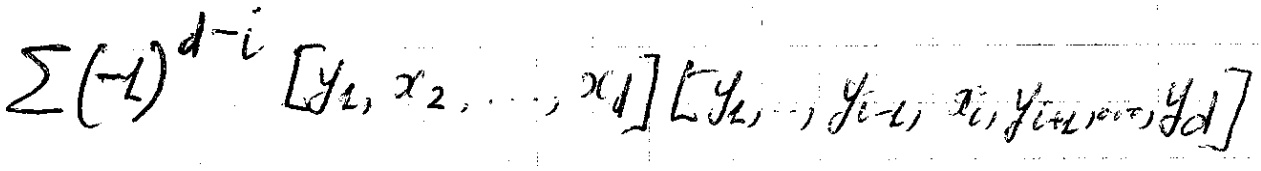


Original source details: GELFAND-P001412

Uncertain reading. The surname tentatively reads «Кунтом», possibly referring to Knuth.

Problem: prove that

$$(6)\quad [x_1,\dots,x_d][y_1,\dots,y_d] = \sum (-1)^{d-i}[y_1,x_2,\dots,x_d] \\ [y_1,\dots,y_{i-1},x_i,y_{i+1},\dots,y_d]$$

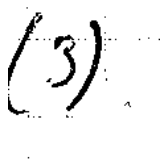


Original source details: GELFAND-P001412

Uncertain reading. The sign $(-1)^{d-i}$ and the indices y_1, x_i follow the manuscript.

$[x_1,\dots,x_d]$ is the determinant formed from the vectors.

(7) Remark: if $[x_1,\dots,x_d]$, $[y_1,\dots,y_d]$ are bases, then the right-hand side has a nonzero term, which gives the exchange axiom (3)



Uncertain reading. The reference (3) follows the manuscript.

Original source details: GELFAND-P001412

(8) Problem (tedious). Hilbert's chain of syzygies:

$$(\text{ring}) \longmapsto (\text{generators}) \longmapsto (\text{relations among the generators})$$
$$\text{— determinants}$$



Uncertain reading. The number has been overwritten and reads as 8.

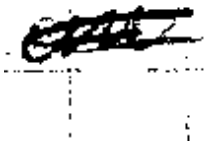
Original source details: GELFAND-P001412

(9) (Claim: (6) gives all the relations among the generators)

(10) a) Write down the whole chain!

b) Discretize the entire chain (write down the combinatorial relations).

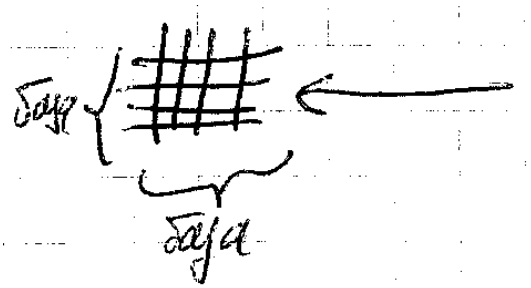
(11) Problem. A bimatroid is a list of minors $\longrightarrow$ [Crossed out: illegible] 2 matroids (rows and columns)



Undeciphered passage. The crossed-out writing before "2 matroids" is illegible.

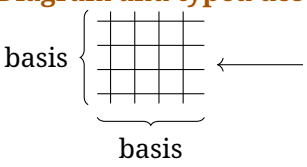
Original source details: GELFAND-P001412

A bimatroid is constructed from 2 matroids



Original source details: GELFAND-P001412

Diagram and typed description.



by def. basis.



Original source details: GELFAND-P001412

Uncertain reading. The word after "by def." is underlined and reads "basis."

But this is not necessary.

There is more information in a bimatroid.

What additional information is there in a bimatroid?

This is an important cohomological concept.

Remark. Algebraic topology is a small island in the usual combinatorial technique.

GELFAND-P001413

30/XI-87 Gelfand Seminar.

I. M. Gelfand. Problem: find the characters of the symmetric group.

(following Frobenius).

On the paper by Kerov: «Realization of representations of the Hecke algebra etc.»

Hint: these are characters of the symmetric group.

The Hecke algebra is the algebra with generators $\{e_k\}$ [above the line: $k = 1, \dots, N-1$] and relations

$$(1) \quad \begin{cases} e_k e_{k+1} e_k - \tau e_k = e_{k+1} e_k e_{k+1} - \tau e_{k+1} \\ e_{k+j} e_k = e_k e_{k+j}, \quad j \geqslant 2, \\ e_k^2 = e_k \end{cases}$$

Other generators

$$(2) \quad \sigma_k = (q+1) e_k - 1, \qquad \tau = \frac{q}{(q+1)^2}$$

This algebra is denoted by $H_N(q)$.

Exercise: write down the relations for σ_k.

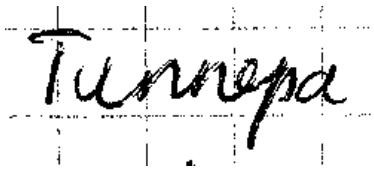
(3) For $q = 1$ (or $\tau = \frac{1}{4}$) we obtain the group algebra of the symmetric group.

(4) Problem: $\dim H_N(q) = N!$

[In the left margin: ⓘ] In the paper by Tipper and James

Proc. London Math. Soc. 1986, v. 52, p. 20–52

recommended!!



Original source details: GELFAND-P001413

Uncertain reading. The first name appears as «Типпера» (Tipper) in the manuscript. The cited publication gives R. Dipper.

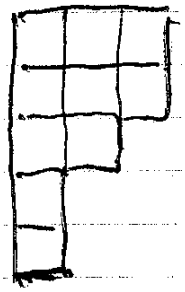
it is proved that the representation theory of $H_N(q)$ is analogous to the modular representation theory of the symmetric group.

This works even in the complex case (see the paper by Bernsh. — Gelf. — Gelf. on Verma modules).

Here it is assumed that $q \neq \sqrt[m]{1}$

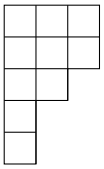
Construction of representations.

Young diagram $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$



Original source details: GELFAND-P001413

Diagram and typed description. (5)



$\sum \lambda_j = |\lambda|$

$\lambda_{i+1} \leqslant \lambda_i$

(λ_i is the number of boxes in the ith column)

This corresponds to a conjugacy class of elements.

GELFAND-P001414

[source number: 52]

Young tableau:

$$(6)\quad t = \{\lambda^{(k)}\}_{k=1}^{N} \quad \text{is a collection of diagrams in which } |\lambda^{(k)}| = k.$$

The usual definition (a different one): the tableau is filled with the numbers $1, 2, \ldots, |\lambda|$, with

$\rightarrow$
$\downarrow$ entries increasing in these directions

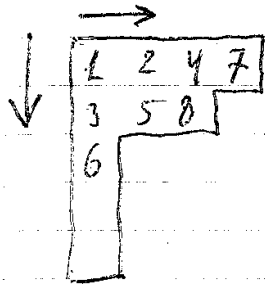
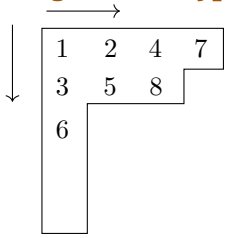


Diagram and typed description.



Original source details: GELFAND-P001414

In (6) they must increase

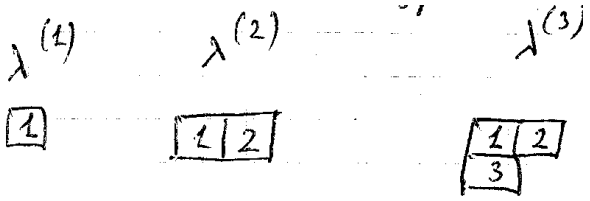
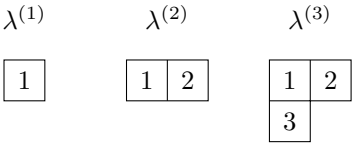


Diagram and typed description.



Original source details: GELFAND-P001414

$$(7)\quad r_k(t) = (i_{k+1} - j_{k+1}) - (i_k - j_k)$$

point k has coordinates (i_k, j_k)

For the diagram:

$$(8)\quad h(\lambda) = r_\lambda + 1$$

$$(9)\quad r_\lambda = \max_{t \in T(\lambda)} \max_{1 \le k \le N} |r_k(t)|,$$

where $T(\lambda)$ is the set of tableaux constructed from the diagram λ, i.e., of shape λ.



Uncertain reading. The manuscript writes the upper bound as $k \le N$.

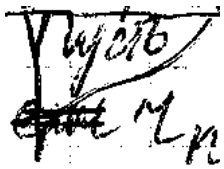
Original source details: GELFAND-P001414

Let $s = (\mu^{(k)})_{k=1}^{N}$, $\quad t = (\lambda^{(k)})_{k=1}^{N}$.

(10) Def. $\quad s \underset{n}{\approx} t$, if $\lambda^{(k)} = \mu^{(k)}$, $k \neq n$

Claim.

(11) [Crossed out: if?] [Inserted above the line: Let] $r_n(t) = \pm 1$. Then $s \underset{n}{\approx} t \Rightarrow s = t$.



Original source details: GELFAND-P001414

Uncertain reading. The crossed-out word tentatively reads "if." The insertion above the line reads "Let."

(12) Let $|r_n(t)| > 1$. Then $\exists!\ s:\ s \underset{n}{\approx} t,\ s \neq t$.

(13) $\quad S_{\lambda}^{(k)} = \|S_{s,t}^{(k)}(\tau)\|_{s,t \in T(\lambda)}$ is a matrix.



Original source details: GELFAND-P001414

Uncertain reading. An unclear mark follows the matrix index, possibly q or a corrected comma.

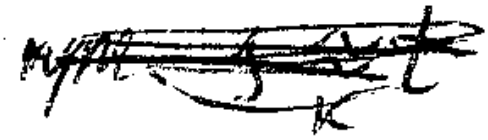
(14) $\quad S_{s,t}^{(k)}(\tau) = \left(W_{r_k(s)}(\tau)\ W_{r_k(t)}(\tau)\right)^{1/2}$ if $s \underset{k}{\approx} t$



Original source details: GELFAND-P001414

Undeciphered passage. There is a blotted mark after the first factor.

[Crossed out: if $s \underset{k}{\approx} t$?] and $S_{s,t}^{(k)}(\tau) = 0$ otherwise.

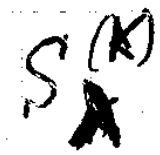


Original source details: GELFAND-P001414

Uncertain reading. The crossed-out condition tentatively reads "if $s \underset{k}{\approx} t$."

GELFAND-P001415

(15) Theorem. Let λ be a diagram with N elements and $q \neq \sqrt[m]{1}$. Then $S_{\lambda}^{(k)}$ give a representation of the Hecke algebra (in the generators e_k).



Original source details: GELFAND-P001415

Uncertain reading. The subscript of $S^{(k)}$ is heavily retraced and reads as λ.

Let us construct the functions W.

$W_{-r}(\tau)$ for $r = 1, 2, \ldots$ is determined from the relations

$$(16) \quad W_{-(r+1)}(\tau) = 1 - \frac{\tau}{W_{-r}(\tau)}, \qquad r \geqslant 0,$$

[Crossed out: and for $r \geqslant 0$?]



Uncertain reading. The crossed-out sign tentatively reads $\geqslant$.

Original source details: GELFAND-P001415

$$(17) \quad W_r(\tau) = 1 - W_{-r}(\tau), \qquad r \geqslant 0;$$

Or:

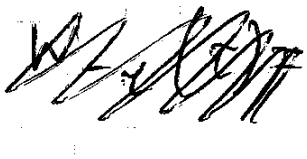
$W_{-r}(\tau)$ is determined [Crossed out: illegible] from its continued fraction expansion:



Undeciphered passage. The crossed-out word after "is determined" is illegible.

Original source details: GELFAND-P001415

$$(18) \quad \text{[Crossed out: } W_{-r}(\tau) = \text{]} \quad 1 - \cfrac{\tau}{1 - \cfrac{\tau}{1 - \ldots}}$$

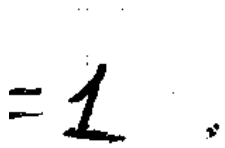


Uncertain reading. The crossed-out expression before the fraction tentatively reads $W_{-r}(\tau) =$.

Original source details: GELFAND-P001415

There is an equation:

$$(19) \quad W_r(\tau) W_{1-r}(\tau) = 1.$$



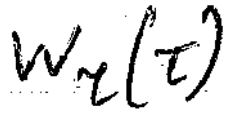
Uncertain reading. The right-hand side of (19) reads 1.

Original source details: GELFAND-P001415

General formula:

$$(20)\quad W_r(\tau) = \frac{q(q^{r-1}-1)}{(q+1)(q^r-1)}, \qquad r = 1, 2, \dots$$

$$(21)\quad W_{-r}(\tau) = \frac{q^{r+1}-1}{(q+1)(q^r-1)}.$$



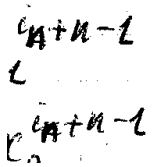
Uncertain reading. The subscript of W reads as r.

Original source details: GELFAND-P001415

(22) Problem: find the characters of the Hecke algebra.

Consider

$$(23)\quad \frac{\begin{vmatrix} x_1^{i_1} & x_1^{i_2+1} & \dots & x_1^{i_n+n-1} \\ x_2^{i_1} & x_2^{i_2+1} & \dots & x_2^{i_n+n-1} \\ \dots & \dots & \dots & \dots \\ x_n^{i_1} & & \dots & x_n^{i_n+n-1} \end{vmatrix}}{\begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & & \\ & & & & \\ 1 & x_n & x_n^2 & & x_n^{n-1} \end{vmatrix}} = \sum a(i,j) P_{j_1} P_{j_2} \dots P_{j_n}$$



Uncertain reading. The subscripts on i in the first two rows of the last column are overwritten. The exponent reads as $i_n + n - 1$.

Original source details: GELFAND-P001415

GELFAND-P001416

[source number: 53]

$j = (j_1, \dots, j_n)$, P_j is a power sum.

Then

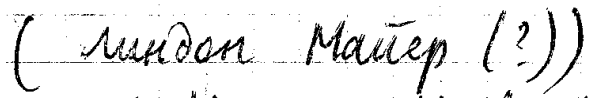
(24) $a(i,j)$ is the character value [inserted above the line: of the symmetric group] on an element with cycle lengths $j_1, \dots, j_n$

(up to a simple constant)

For the symmetric group, see Macdonald's book "Symmetric Functions and Hall Polynomials."

Location of the poles of Ruelle's zeta function.

(Lyndon Mayer (?))



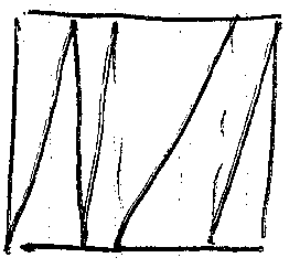
Uncertain reading. The manuscript appears to read "Lyndon Mayer (?)". The cited article is by Dieter H. Mayer.

Original source details: GELFAND-P001416

Letters in Math. Physics, $14:2$, 1987.

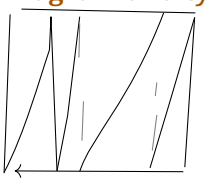
I is an interval, $T : I \to I$ is piecewise linear

$$I = \bigcup_j I_j, \qquad T(I_j) = I.$$



Original source details: GELFAND-P001416

Diagram and typed description.



The attachment of the faint strokes to the branches and the exact locations of the breaks are unclear.

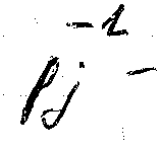
$$(1) \quad \zeta_T(z,\varphi) = \frac{\det(1 - zL_1)}{\det(1 - zL_2)},$$

where

$$L_1 f(z) = \varphi(z) \sum \frac{1}{\rho_j} f(\psi_j(z)),$$

$$L_2 f(z) = \varphi(z) \sum f(\psi_j(z)),$$

ψ_j are the local inverses of T on the interval I_j, and ρ_j^{-1} is the slope.

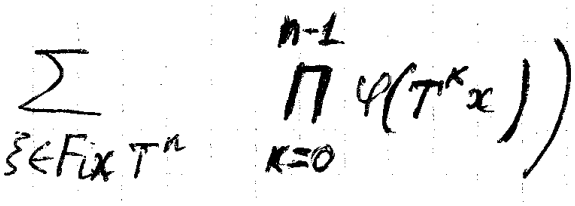


Original source details: GELFAND-P001416

Uncertain reading. The slope letter can be read as ρ_j or p_j.

In the general case:

$$(2) \quad \zeta_T(z,\varphi) = \exp\left(\sum_{n=1}^{\infty} \frac{z^n}{n} \sum_{\xi \in \operatorname{Fix} T^n} \prod_{k=0}^{n-1} \varphi(T^k x)\right)$$



Original source details: GELFAND-P001416

Uncertain reading. The second sum is indexed by $\xi \in \operatorname{Fix} T^n$. The product uses x.

GELFAND-P001417

7/XII-87 Gelfand Seminar. Some remarks on the paper by

D. Mayer Lett. in Math. Phys. 1987, 14:2

On Ruelle's ζ function

F is the set of spin values (finite).

We study spin configurations on a lattice

$$(\xi_i), \qquad -\infty < i < \infty, \qquad \xi_i \in F.$$

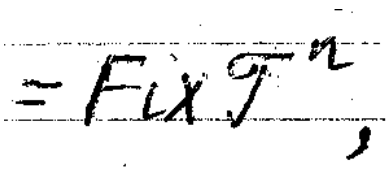
$M \subseteq F \times F$ is the admissible set. We require

$$(1) \quad (\xi_i, \xi_{i+1}) \in M$$

The simplest example of a Ruelle ζ function is the Artin–Mazur ζ function.

Consider the set of periodic configurations and their number:

$$(2) \quad N_n = \#\{(\xi_i) \mid \mathcal{T}^n(\xi_i) = (\xi_i)\} = \operatorname{Fix} \mathcal{T}^n,$$



Original source details: GELFAND-P001417

Uncertain reading. The last term of (2) is written as $\operatorname{Fix} \mathcal{T}^n$ without a cardinality sign.

where $\mathcal{T}$ is the transformation on the space of sequences

$$\mathcal{T}(\xi_i) = \xi_{i+1}$$

(shift)

Ruelle's ζ function:

$$(3) \quad \zeta_{\mathcal{T}}(s) = \exp\left(\sum_{n=1}^{\infty} \frac{N_n}{n} s^n\right)$$

the series converges for $|s| < \dfrac{1}{|F|}$.

Does an analytic continuation exist? Consider the matrix



Original source details: GELFAND-P001417

Uncertain reading. The end of the first word and the initial letter of "analytic" are retraced in the manuscript.

$$(4) \quad T = (T_{\alpha\beta}) = \begin{cases} 1, & (\alpha, \beta) \in M, \\ 0, & (\alpha, \beta) \notin M. \end{cases}$$

Then

$$(5) \quad \zeta_{\mathcal{T}}(s) = \frac{1}{\det(1 - sT)}$$

This shows that $\zeta_{\mathcal{T}}(s)$ is meromorphic.

The radius of convergence may exceed $1/|F|$.

For the proof, observe that

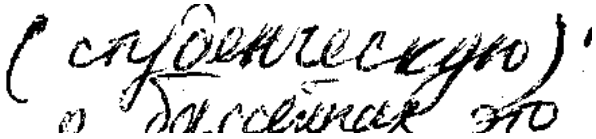
$$N_n = \operatorname{Tr} T^n. \tag{6}$$

GELFAND-P001418

[source number: 62]

Remark by I. M. G.: this resembles the work of Dynkin and Dmitriev:

consider a stochastic matrix (p_{ik}) [inserted above the line: (a student's?)] (in the problem about basins this is the [crossed-out sign: illegible] fraction of the water that flows from basin $i$to basin k).



Original source details: GELFAND-P001418

Uncertain reading. The insertion above the line tentatively reads «студенческую» (student's).



Original source details: GELFAND-P001418

Undeciphered passage. The sign before "fraction" is crossed out and illegible.

$$\underline{P} = (p_{ik}), \qquad \sum_k p_{ik} = 1.$$

This depends on the eigenvalues of P.

a) The eigenvalues lie in the disk $|\lambda| \leq 1$,

b) 1 is an eigenvalue.

What region do the eigenvalues fill?

It is the interior of some algebraic curve.

They went about it the wrong way.

There is a whole set of eigenvalues!!

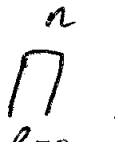
and one should plot the set of eigenvalues in $\mathbb{C}^n$ — it ought to be a beautiful region (Problem!!)

The projection onto a single axis can be complicated!

[Double line in the left margin.] In Ruelle's problem it is interesting to look at all the eigenvalues of the matrix T.

Speaker. Generalization of example (3)

$$(7) \quad \zeta_{\mathcal{T}}(s, \varphi) = \exp\left(\sum_{n=1}^{\infty} \frac{s^n}{n} \sum_{\xi \in \mathrm{Fix}(\mathcal{T}^n)} \prod_{\ell=0}^{n} \varphi(\mathcal{T}^\ell \xi)\right)$$



Uncertain reading. The upper limit of the product in (7) is written as n.

Original source details: GELFAND-P001418

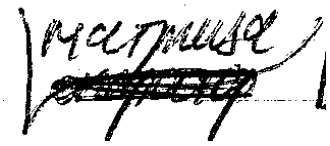
Here

$$\underset{\text{function}}{\varphi} = \varphi(\xi_1, \ldots, \xi_k), \qquad \xi_j \in F$$

(φ depends only on the first k values).

((3) is obtained for $\varphi = 1$)

(8) Claim. $\exists$ [Crossed out: operator?] [inserted above the line: matrix] L (finite-dimensional), such that:



Uncertain reading. The crossed-out word tentatively reads "operator." The insertion above it reads "matrix."

Original source details: GELFAND-P001418

$$\zeta_T(s,\varphi) = \frac{1}{\det(1-sL)} \tag{9}$$

Specifically, consider

$$F' = F^k;$$

[Crossed out: illegible]

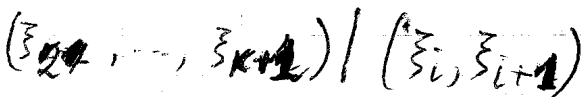


Original source details: GELFAND-P001418

Undeciphered passage. The writing before M' is crossed out and illegible.

$$M' \subset F' \times F'$$

$$M' = \{(\xi_1,\dots,\xi_k);(\xi_2,\dots,\xi_{k+1}) \mid (\xi_i,\xi_{i+1}) \in M\} \tag{10}$$



Original source details: GELFAND-P001418

Uncertain reading. The subscripts 2 and $k+1$ in the second tuple are overwritten. The $i+1$ in the condition is also retraced.

Then take

$$\ell_{\alpha\beta} = \begin{cases} \varphi(\alpha), & (\alpha,\beta) \in M', \\ 0, & (\alpha,\beta) \notin M', \end{cases} \qquad L = (\ell_{\alpha\beta}).$$

GELFAND-P001419

I. M. G.

NB: one needs to know the p-adic numbers

(be able to add, multiply, and divide them, [crossed out: illegible] and extract roots)



Original source details: GELFAND-P001419

Undeciphered passage. The mark after "divide" is crossed out and illegible.

Beilinson A. p-adic numbers and physics.

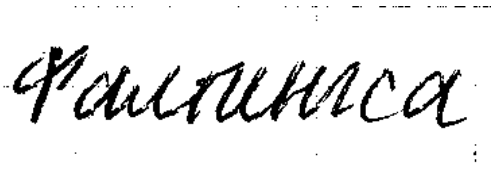
On some of Manin's p-adic ideas in physics!



Original source details: GELFAND-P001419

Uncertain reading. The last mark resembles "!" and may read "?".

1°. Witten listened to the talk by Faltings with great interest and then bought all the books on number theory. The only thing he understood from the talk was the formula



Uncertain reading. The surname in the manuscript tentatively reads «Фалтингса» (Faltings).

Original source details: GELFAND-P001419

$$(1)\quad |a|_\infty \cdot |a|_2|a|_3 \ldots = 1, \qquad a \in \mathbb{Q}.$$

For $\frac{1}{2}$ we have here:

$$\frac{1}{2} \cdot 2 \cdot 1 \ldots = 1.$$

There are 3 papers:

1) Volovich (in Russian)

2) Yu. I. Manin (in English)

3) Witten and someone else



Uncertain reading. An isolated illegible mark stands to the right of the third entry.

Original source details: GELFAND-P001419

All 3 papers appeared in September.

Witten's text.

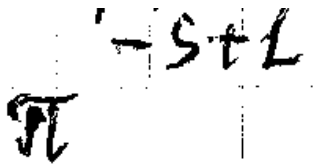
Consider the functional equation for the Riemann ζ function

$$(2)\quad \pi^{-s/2}\Gamma(s/2)\prod(1-p^{-s})^{-1} = \pi^{-(1-s)/2}\Gamma\left(\frac{1-s}{2}\right)\prod(1-p^{-(1-s)})^{-1}.$$

(analytic continuation is understood)

or (formally):

$$(3)\quad \frac{\pi^{-s+1}\Gamma(s/2)}{\Gamma\left(\frac{1-s}{2}\right)} = \prod_p \frac{1-p^{-s}}{1-p^{s-1}}.$$



Uncertain reading. The exponent of π is written as $-s+1$.

Original source details: GELFAND-P001419

On the left ≈ the Veneziano–Witten amplitude for an open string [inserted above the line: (with momentum s)]. On the right is a divergent expression, but what matters is that it is written as a product over the primes!

GELFAND-P001420

[source number: 63]

2°. On Manin's work

(1) Claim: The Tamagawa measure $= 1$.

In more detail:

$$\mathrm{SL}(n,\mathbb{R}) \supset \mathrm{SL}(n,\mathbb{Z}).$$

On $\mathrm{SL}(n,\mathbb{R})$ there is an invariant [inserted above "invariant": «left»] measure (the standard one).



Undeciphered passage. A crossed-out mark is visible below the end of "measure."

Original source details: GELFAND-P001420

Normalize [crossed out: illegible] using the integer matrices.



Undeciphered passage. The writing after "Normalize" is crossed out and illegible.

Original source details: GELFAND-P001420

$$\mathrm{vol}\big(\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z})\big) - ?$$

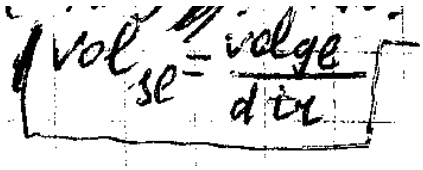
[In the right margin:] (it is constructed from the standard basis in the Lie algebra $\mathfrak{sl}(n,\mathbb{R})$ or in $\mathfrak{gl}(n,\mathbb{R})$ [crossed out: illegible] standard basis.



Undeciphered passage. The mark after $\mathfrak{gl}(n,\mathbb{R})$ is crossed out and illegible.

Original source details: GELFAND-P001420

$$\boxed{\mathrm{vol}_{sl} = \frac{\mathrm{vol}_{gl}}{d\ \mathrm{tr}}}$$



Uncertain reading. The small letters next to "vol" read sl and gl.

Original source details: GELFAND-P001420

$\mathrm{SL}(2,\mathbb{R})$ consists of conformal transformations of the unit disk, acting transitively on S^*H^2. The invariant volume on S^*H^2 is [crossed out: illegible] $\mathrm{vol}_{H^2} \otimes d\varphi$.



Undeciphered passage. The mark before vol_{H^2} is crossed out and illegible.

Original source details: GELFAND-P001420



Uncertain reading. The letter after d reads as φ.

Original source details: GELFAND-P001420

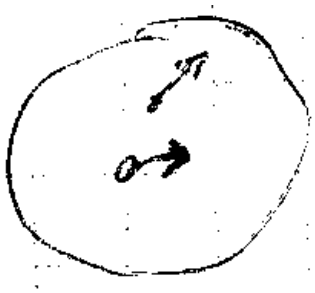
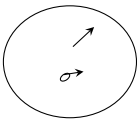


Diagram and typed description.



Original source details: GELFAND-P001420

Or take the quotient by the maximal compact subgroup

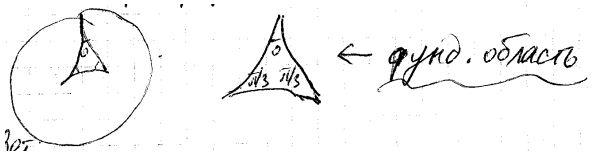
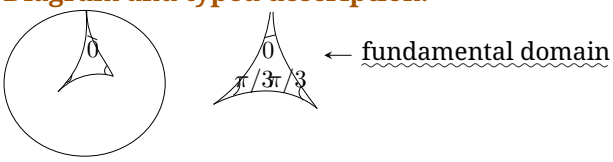


Diagram and typed description.



Original source details: GELFAND-P001420

Answer.

$$(2)\quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z}) = \zeta(2)\zeta(3)\ldots\zeta(n)$$

$\mathbb{A}$ denotes the adeles.

$$(3)\quad \mathrm{SL}(n,\mathbb{A}) \sim \mathrm{SL}(n,\mathbb{R}) \times \mathrm{SL}(n,\mathbb{Q}_2) \times \ldots$$
$$\cup$$
$$\mathrm{SL}(n,\mathbb{Q})$$

$$(4)\quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{A})/\mathrm{SL}(n,\mathbb{Q}) = 1.$$

$$\mathrm{vol}\,\mathrm{SL}(n,\mathbb{A})/\mathrm{SL}(n,\mathbb{Q}) \sim \mathrm{vol}\,\mathrm{SL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{Z})$$
$$\prod \mathrm{vol}\,\mathrm{SL}(n,\mathbb{Z}_p).$$

$$(5)\quad \mathrm{vol}\,\mathrm{SL}(n,\mathbb{Z}_p) \sim \#\mathrm{SL}_n(\mathbb{F}_p)/p^{n^2-1}.$$

Exercise: compute $\#\mathrm{SL}_n(\mathbb{F}_p)$.

GELFAND-P001421

(6) Number of elements of $\mathrm{GL}(n, \mathbb{F}_p)$

$$(p^n - 1)(p^n - p)(p^n - p^2) \ldots = p^{n^2}(1 - p^{-n})(1 - p^{-n+1}) \ldots (1 - p^{-1}).$$

(7) For $\mathrm{SL}(n, \mathbb{F}_p)$ divide by $p - 1$:

$$p^{n^2-1}(1 - p^{-n})(1 - p^{-n+1}) \ldots (1 - p^{-2}).$$

This is a product of Euler factors.

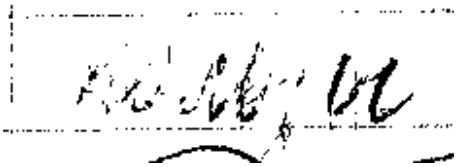
(8) Yu. I. Manin's conjecture: an analogue of formula (2) can be written for the moduli spaces of curves arising in string theory.

I. M. G.

Mathematical question: the volume of Teichmüller space has a p-adic analogue.

Or: what is adelic Teichmüller space

An example of the power of Witten's work:



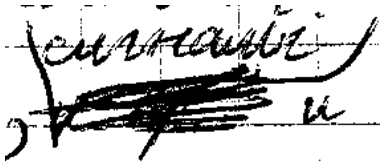
Original source details: GELFAND-P001421

Uncertain reading. The faint word between "example" and "work" may read «мощи» or «силы» (power).

Laumon did work in number theory based on Witten's ideas from Morse theory.

Idea: consider a function $u \in \mathcal{E}'(\mathbb{R})$, $\widetilde{u}(\xi) \in C^\infty$ and all the singularities will go into the asymptotics at infinity.

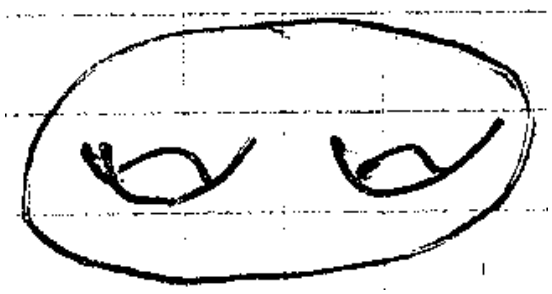
Advice from I. M. G.: read Kharkevich's book «Spectra, [crossed out: illegible] [inserted above the line: Signals] and Noise», to understand the Fourier transform.



Original source details: GELFAND-P001421

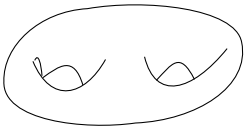
Undeciphered passage. The writing in the book title is crossed out. "Signals" is inserted above it.

On the work of Laumon.



Original source details: GELFAND-P001421

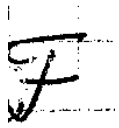
Diagram and typed description.



The short stroke at the left of the first loop follows the visible outline. Its purpose is unclear.

Consider a surface $\mathcal{F}$

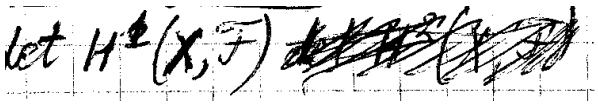
$$H^i(X, \mathcal{F}), \qquad i = 0, 1, 2.$$



Uncertain reading. After “surface” the manuscript writes $\mathcal{F}$, and then $H^i(X, \mathcal{F})$.

Original source details: GELFAND-P001421

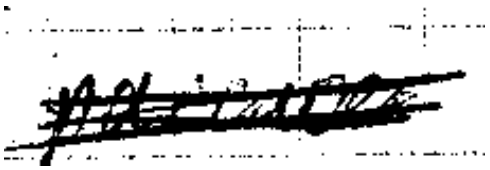
$$(1) \quad \det H^\bullet(X, \mathcal{F}) = \frac{\det H^0(X, \mathcal{F}) \det H^2(X, \mathcal{F})}{\det H^1(X, \mathcal{F}) \quad \text{[crossed out: } \det H^2(X, \mathcal{F})\text{]}}$$



Uncertain reading. The first superscript in the denominator is retraced and reads 1. The crossed-out factor tentatively reads $\det H^2(X, \mathcal{F})$.

Original source details: GELFAND-P001421

([crossed out: finite?] vector spaces. det is the top exterior power)



Uncertain reading. The crossed-out word before “vector spaces” tentatively reads “finite.”

Original source details: GELFAND-P001421

Automorphisms?

Consider a triangulation. $\det H^\bullet$ is identified with the product of the det of the terms of the complex (an Euler-type formula): canonically

GELFAND-P001422

[source number: 64]

$$(2) \quad \det(C^\bullet) = \det(H^\bullet(C))$$

$2 - 2g$ is the Euler characteristic.

On the other hand

$$2g - 2 = \deg \underbrace{K}_{\text{canonical class}}$$

Let

D be a divisor.

$$(3) \quad \det \mathcal{F}_D = \bigotimes (\det \mathcal{F}_{x_i})^{\otimes a_i}, \qquad D = \sum a_i x_i$$

[crossed out: $D \longrightarrow D + K$?]

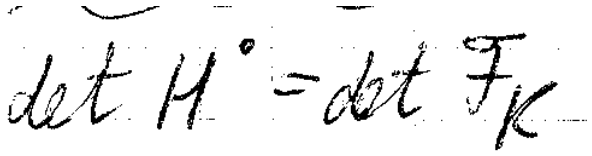


Uncertain reading. The crossed-out expression tentatively reads $D \to D + K$ or $D \mapsto D + K$.

Original source details: GELFAND-P001422

Deligne's formula:

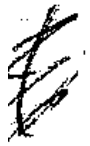
$$(4) \quad \det H^\bullet = \det \mathcal{F}_K$$



Original source details: GELFAND-P001422

Uncertain reading. The right-hand side of (4) reads $\det \mathcal{F}_K$ without the exponent -1.

[At right: crossed out: illegible]



Original source details: GELFAND-P001422

Undeciphered passage. The separate crossed-out mark to the right of the formula is illegible.

(fix the complex structure and the choice of a point in the canonical class)

For an arbitrary constructible sheaf one also computes det and there is an analogue of Deligne's formula!

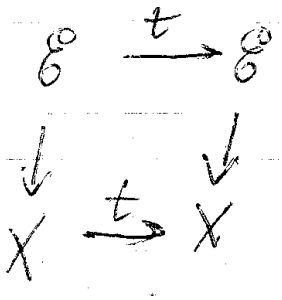
Here on $\mathbb{A}^1$ functions with singularities arise

Klyachko. Toric bundles.

A toric variety is a variety X on which an algebraic torus T, acts, with an orbit isomorphic to T.

$$X = \bigcup_\sigma O_\sigma, \quad \text{is the decomposition into orbits}$$

$\mathcal{E}$ is a bundle over X with a toric structure (and the action lifts! The diagram commutes)



Original source details: GELFAND-P001422

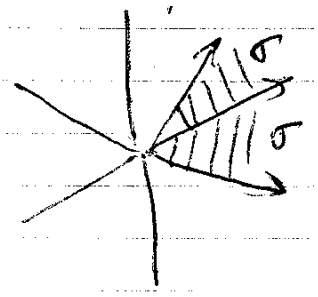
Diagram and typed description.

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{t} & \mathcal{E} \\ \downarrow & & \downarrow \\ X & \xrightarrow{t} & X \end{array}$$

One can describe the moduli of [inserted above the line: «such»] bundles in the language of linear algebra.

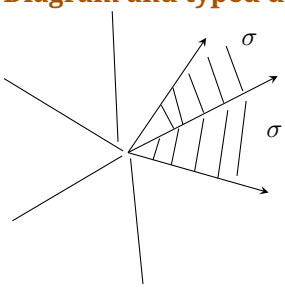
GELFAND-P001423

The structure of X is described by a fan Σ in the Lie algebra of the torus.



Original source details: GELFAND-P001423

Diagram and typed description.



(cones)

$|\Sigma|$ is the set of 1-dimensional cones in the fan Σ.

Theorem. (category of toric bundles) $\Longleftrightarrow$ (category of vector spaces [crossed-out sign: illegible] E with a collection of [inserted above the line: «decreasing $\mathbb{Z}$-indexed»] filtrations $E^\alpha(i)$, $\alpha \in |\Sigma|$, satisfying the condition



Undeciphered passage. The crossed-out mark before E is illegible.

Original source details: GELFAND-P001423

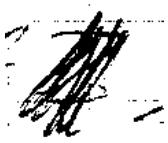
A) The filtrations E^α, $\alpha \in \sigma \in \Sigma$, are distributive.

Problems. 1) Describe the geometry of the bundle (cohomology, characteristic classes, and so on) in terms of the filtrations

2) classification — ?

Let Σ be a simplicial complex. Assign a filtration to a vertex

A representation of it is a collection of filtrations in a vector space [inserted above the line: «E:»] $E^\alpha(i)$, $\alpha \in$ [crossed-out sign: illegible] Σ,



Undeciphered passage. The mark between $\in$ and Σ is crossed out and illegible.

Original source details: GELFAND-P001423

$$E \ldots \supset E^\alpha(i) \supset E^\alpha(i+1) \supset \ldots 0$$

$$\alpha \in (\text{vertices of the complex}) = \Sigma.$$

The spaces $E^\alpha(i)$, $\alpha \in \underset{\text{simplex}}{\sigma}$, are distributive for fixed σ (for all i)

[Outlined insertion: (i.e., they generate a distributive lattice)]

For example, if

$\alpha_1, \alpha_2, \alpha_3$ are the vertices of a triangle, then

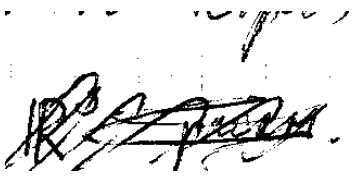
E^{α_1} E^{α_2} E^{α_3} are distributive, i.e.,

$$(E^{\alpha_1} + E^{\alpha_2}) \cap E^{\alpha_3} = E^{\alpha_1} \cap E^{\alpha_3} + E^{\alpha_2} \cap E^{\alpha_3}$$

Classify: a good problem

For a triangle, the solution is trivial

For the surface of a tetrahedron $\Sigma = \partial\Delta$, Δ is a simplex, and the answer is as follows. In [crossed out: an $\mathbb{R}^3$ space?] E, $\dim E = 3$, consider 4 2-dimensional subspaces in general position.



Uncertain reading. The crossed-out passage tentatively reads “an $\mathbb{R}^3$ space.”

Original source details: GELFAND-P001423

$$\mathcal{E} = (E;\ E^\alpha, E^\beta, E^\gamma, E^\delta)$$



Original source details: GELFAND-P001423

Uncertain reading. The last superscript is retraced and reads as δ.

GELFAND-P001424

[source number: 65]

we still need to take [above the line: their?] exterior powers

$$\Lambda^i\mathcal{E} = (\Lambda^i E, \Lambda^i E^\alpha, \Lambda^i E^\beta, \Lambda^i E^\gamma, \Lambda^i E^\delta)$$



Original source details: GELFAND-P001424

Uncertain reading. The insertion above the line tentatively reads «их» (“their”).

A general representation decomposes into a direct sum of 1-dimensional ones and ones of this kind.

I. M. G. The right problem: isolate in $\forall$ problem the solvable pieces. (Gelfand–Ponomarev philosophy)

V. [crossed-out initial: illegible] Kac describes all this in the language of graphs and Kac–Moody algebras: vertex dimensions = root coordinates, etc.



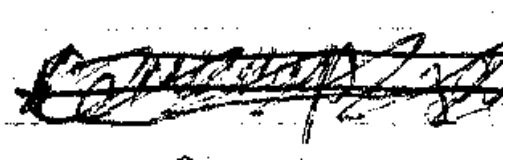
Original source details: GELFAND-P001424

Uncertain reading. The crossed-out initial between “V.” and “Kac” is illegible.

The number of parameters is determined.

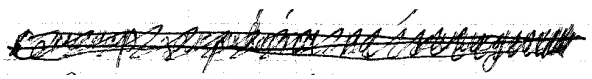
Lect. Notes, 996; an earlier work — Inventiones. (conference talk)

[Crossed out: Seminar? ...]



Original source details: GELFAND-P001424

Uncertain reading. The first word of the crossed-out line tentatively reads “Seminar.”



Original source details: GELFAND-P001424

Undeciphered passage. The rest of the crossed-out line is illegible.

Next time: 8 February.